SEEM Geo

ERT & MASW Processing Suite

Project: Clean Test Project

Author: Test
Institution: Test

Corridor: W → E
Length: 15 km
CRS: UTM
Azimuth: 90° (N90°E)

Acquisition Summary:

ERT: Wenner-alpha — 48 electrodes @ 2.0 m
MASW: 24 geophones @ 2.0 m, 4.5 Hz
Sites: 12 collocated boreholes

Generated: 17 October 2026, 22:57

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Analysis Summary

1. Site Classification Results (RB-301 Site Class)

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MW-01        Vs30 =   290 m/s   →   D (stiff soil)
MW-02        Vs30 =   307 m/s   →   D (stiff soil)
MW-03        Vs30 =   286 m/s   →   D (stiff soil)
MW-04        Vs30 =   323 m/s   →   D (stiff soil)
MW-05        Vs30 =   317 m/s   →   D (stiff soil)
MW-06        Vs30 =   616 m/s   →   C (very dense soil & soft rock)
MW-07        Vs30 =   604 m/s   →   C (very dense soil & soft rock)
MW-08        Vs30 =   638 m/s   →   C (very dense soil & soft rock)
MW-09        Vs30 =   648 m/s   →   C (very dense soil & soft rock)
MW-10        Vs30 =   648 m/s   →   C (very dense soil & soft rock)
MW-11        Vs30 =   282 m/s   →   D (stiff soil)
MW-12        Vs30 =   628 m/s   →   C (very dense soil & soft rock)
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2. Correlation Framework Fits

corr_1_rho_RQD_G1:

n=4 · status: min_df
slope=96.5 intercept=-223 R^2=0.763 p=0.126

⚠ *n=4: minimum df=2; CI wide*

corr_2_Vs_N60:

n=6 · status: ok
slope=3.23 intercept=144 R^2=0.106 p=0.529

corr_3_Vs_RQD_G1:

n=4 · status: min_df
slope=0.21 intercept=-76.1 R^2=0.702 p=0.162

⚠ *n=4: minimum df=2; CI wide*

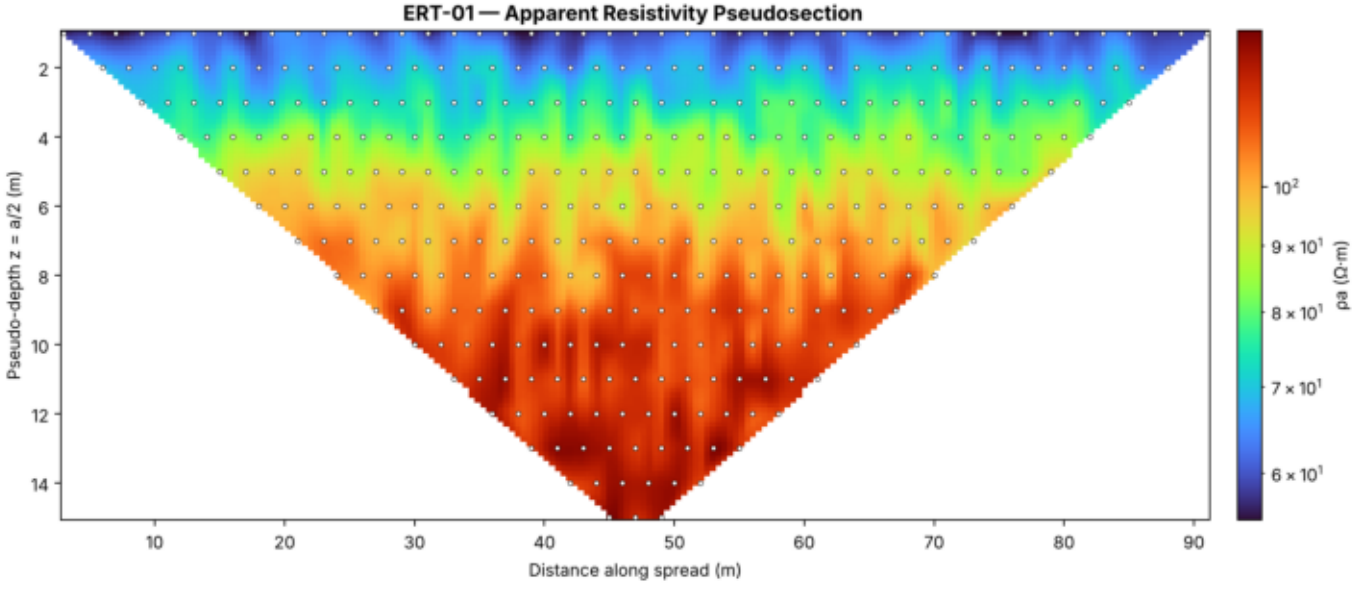
corr_4_Vs_UCS:

n=7 · status: ok
slope=0.00236 intercept=0.223 R^2=0.968 p=6.54e-05

3. Statistical Qualifications (per Chapter V declaration)

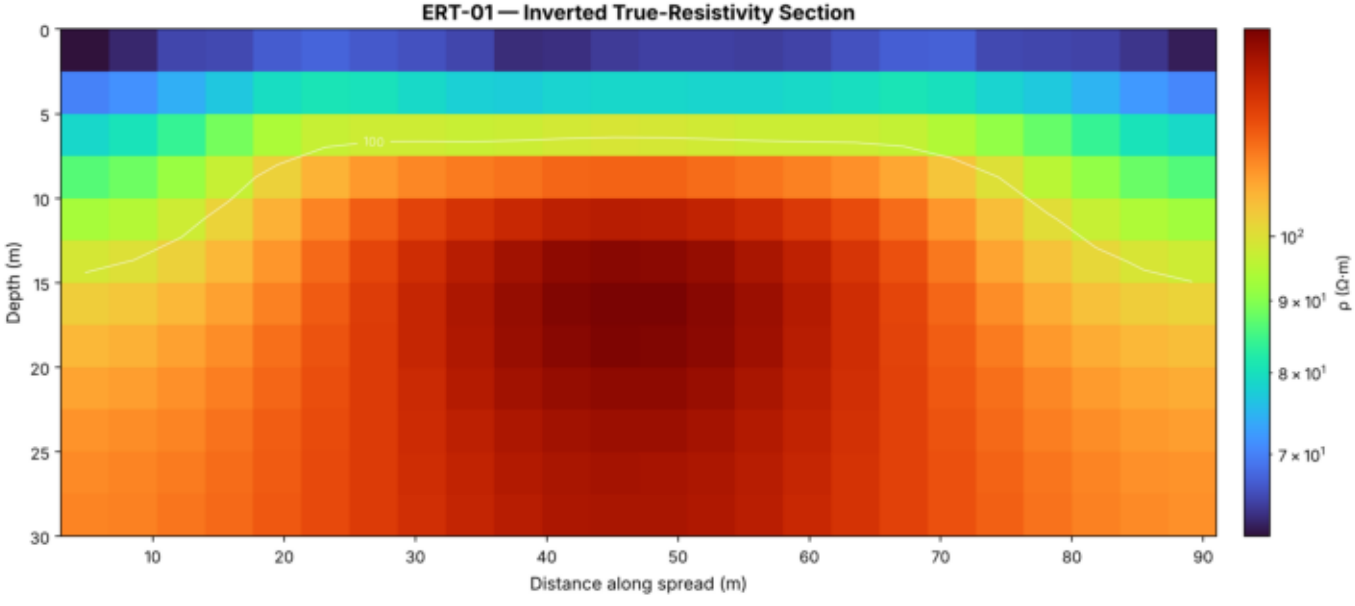
- Group G1 (Schist, n=4): regression at theoretical minimum df=2; CI wide.
- Group G3 (Weak sedimentary, n=3): trend-line level only; R^2 not reliable.
- Group G2 (Granite, n=2): descriptive characterisation only, no inference.
- Group G5 (Residual, n=3): validation targets for the predictive framework.

ERT-01 — Apparent Resistivity Pseudosection (BH1, Group G3)



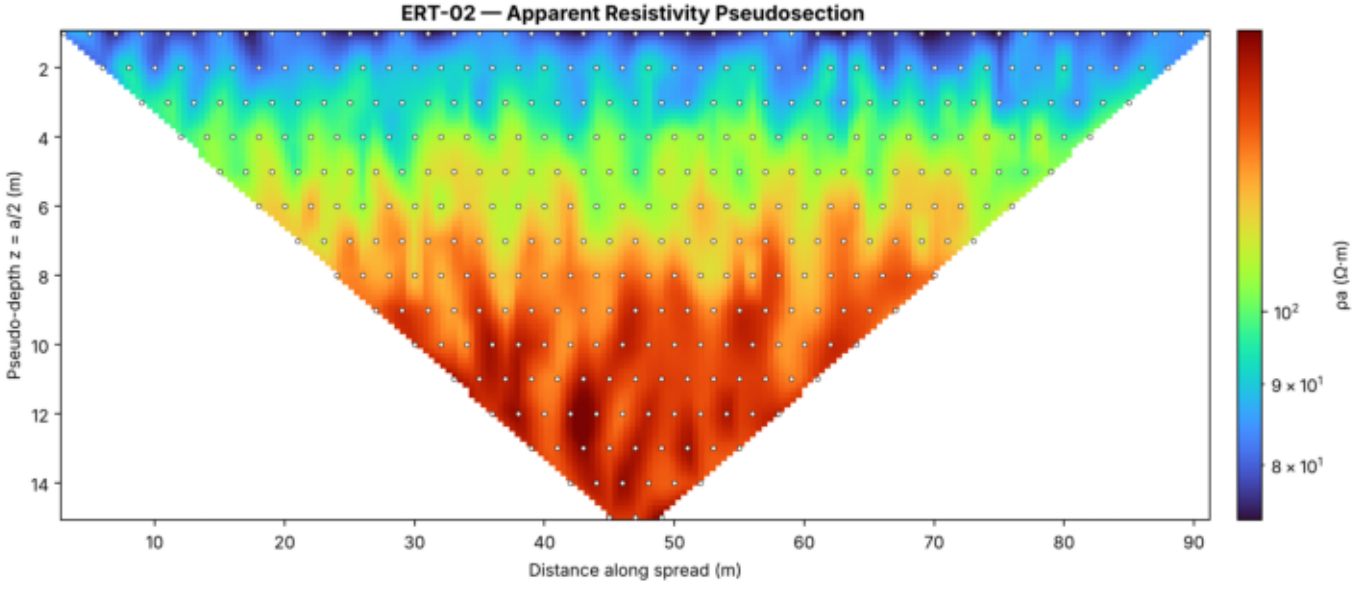
Wenner-alpha, 48 electrodes @ 2 m, spread 100 m, azimuth N67.6°E.

ERT-01 — Inverted True-Resistivity Section (BH1, Group G3)



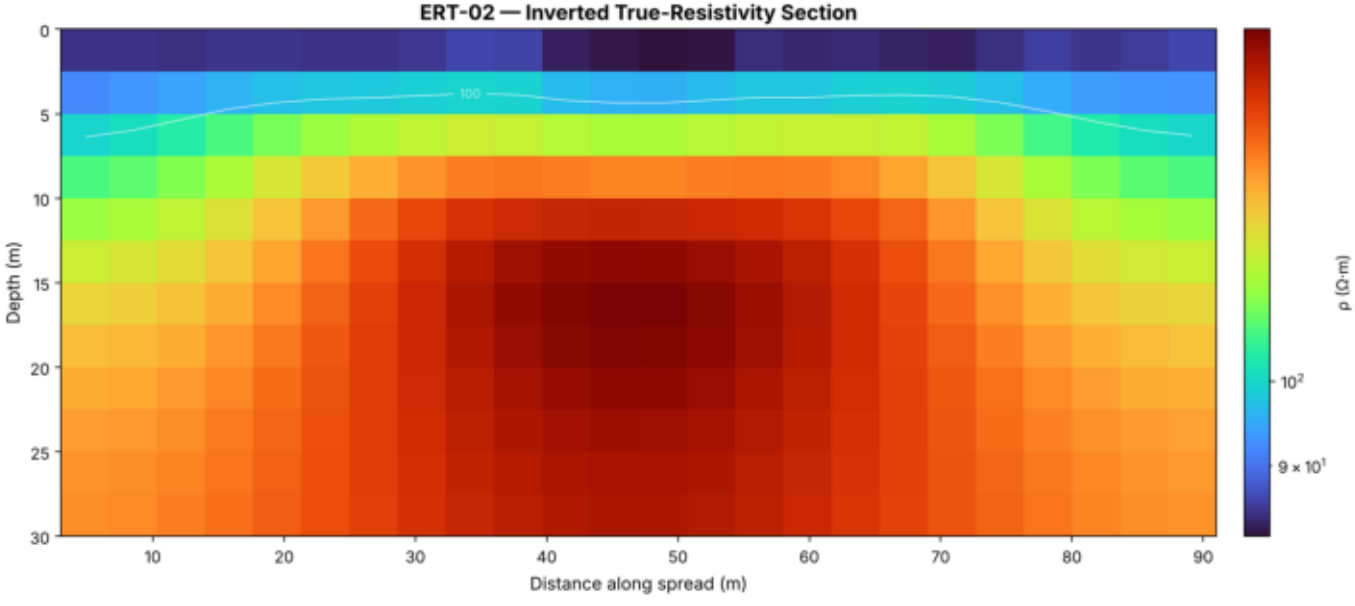
Occam smoothness-constrained least-squares (Loke & Barker 1996).

ERT-02 — Apparent Resistivity Pseudosection (BH2, Group G3)



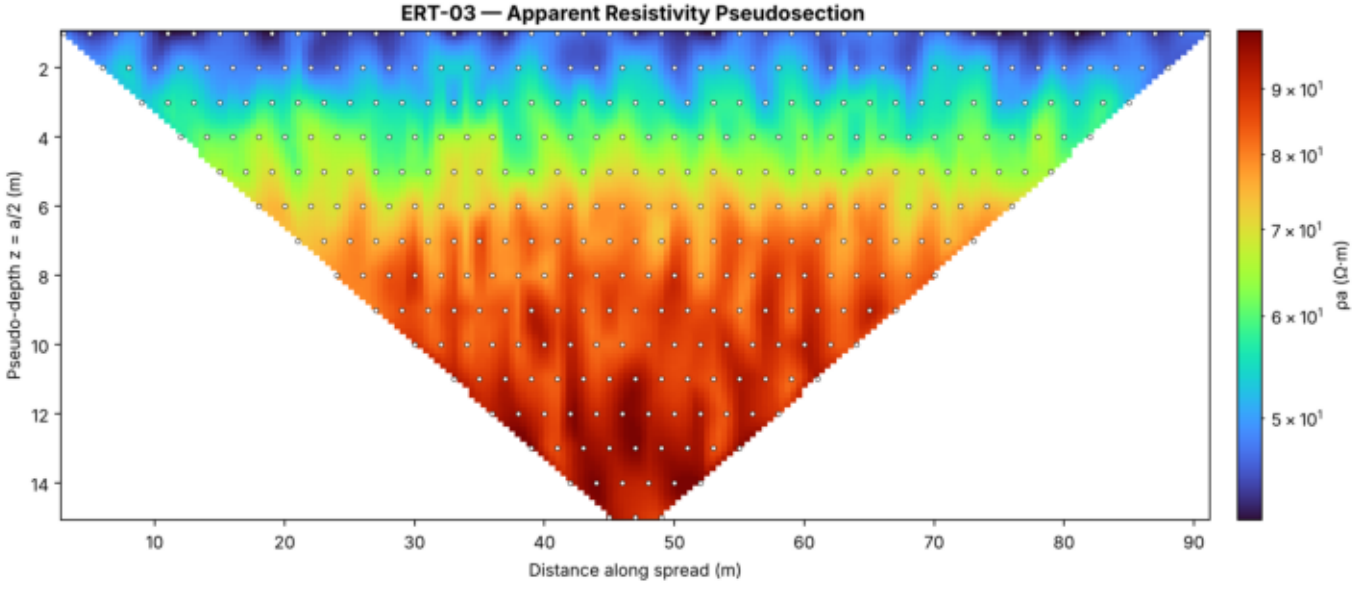
Wenner-alpha, 48 electrodes @ 2 m, spread 100 m, azimuth N67.6°E.

ERT-02 — Inverted True-Resistivity Section (BH2, Group G3)



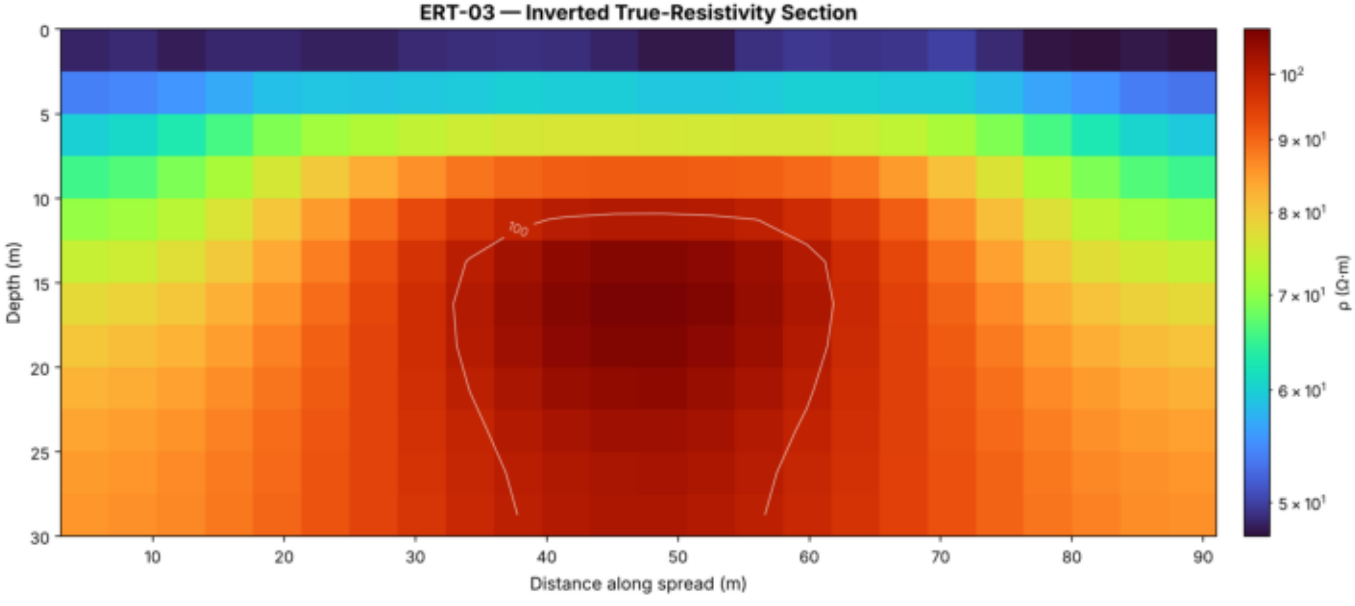
Occam smoothness-constrained least-squares (Loke & Barker 1996).

ERT-03 — Apparent Resistivity Pseudosection (BH3, Group G5)



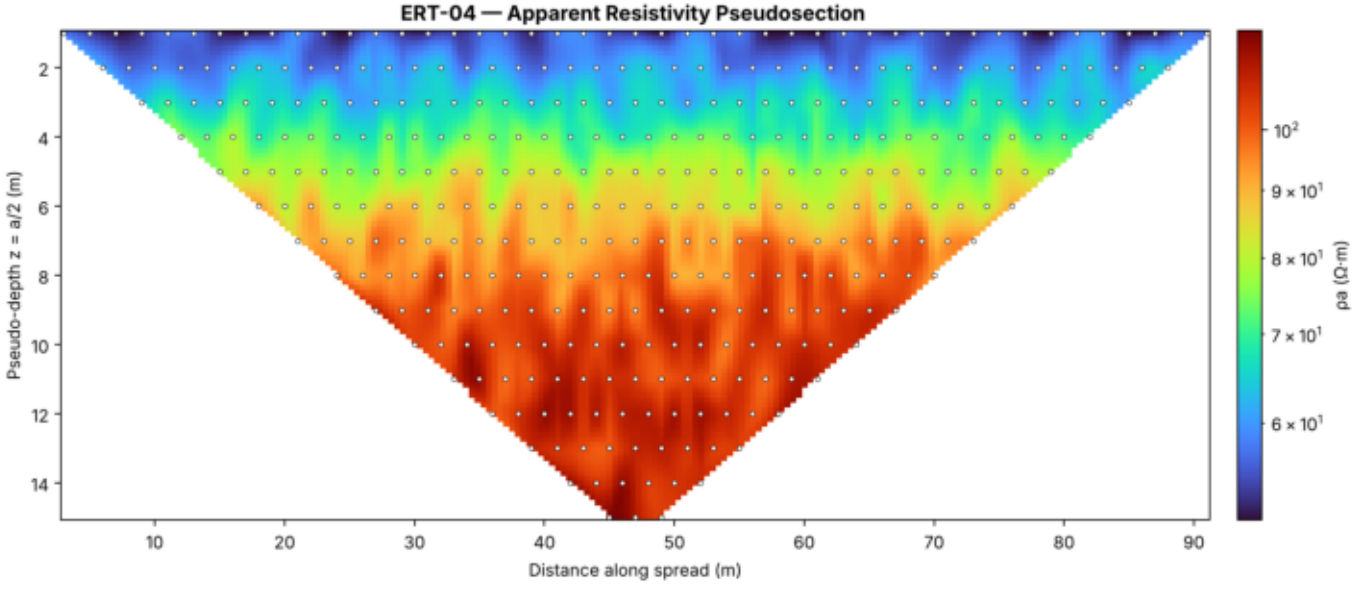
Wenner-alpha, 48 electrodes @ 2 m, spread 100 m, azimuth N67.6°E.

ERT-03 — Inverted True-Resistivity Section (BH3, Group G5)



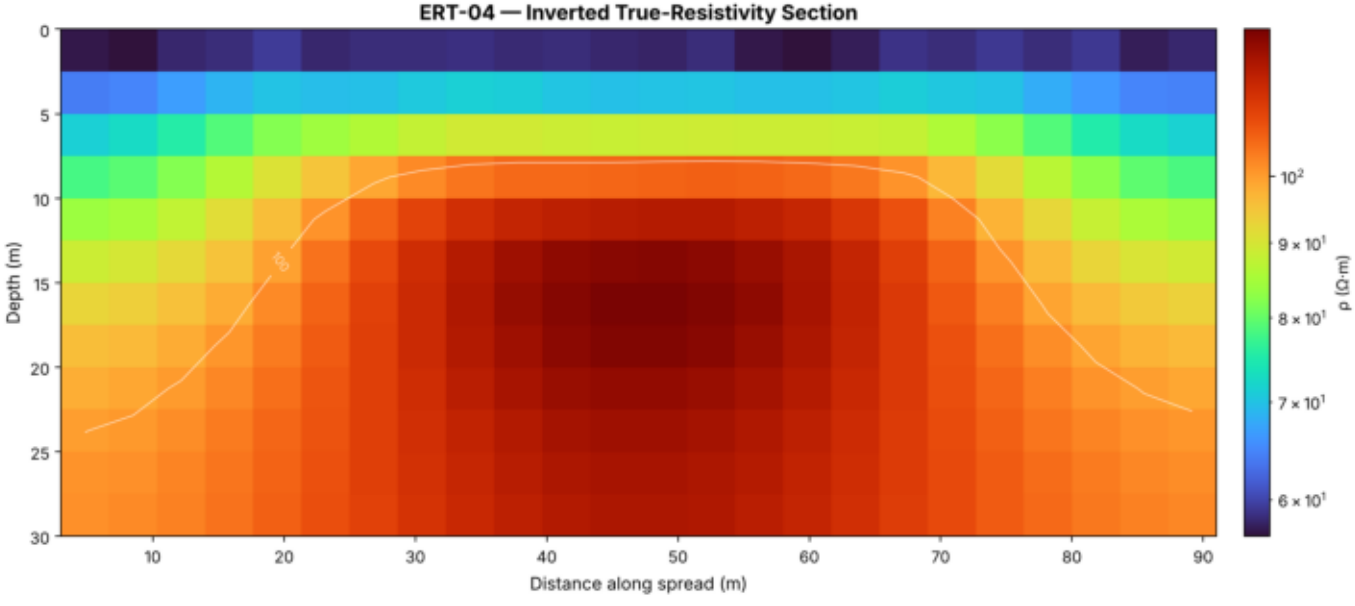
Occam smoothness-constrained least-squares (Loke & Barker 1996).

ERT-04 — Apparent Resistivity Pseudosection (BH4, Group G5)



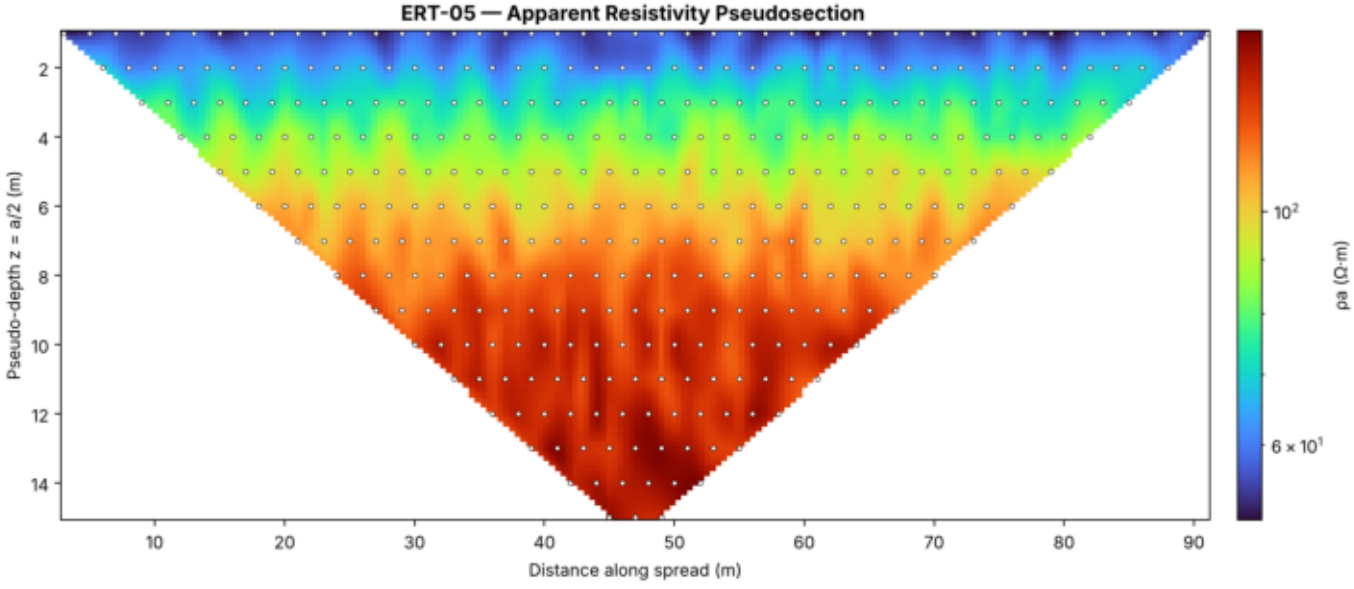
Wenner-alpha, 48 electrodes @ 2 m, spread 100 m, azimuth N67.6°E.

ERT-04 — Inverted True-Resistivity Section (BH4, Group G5)



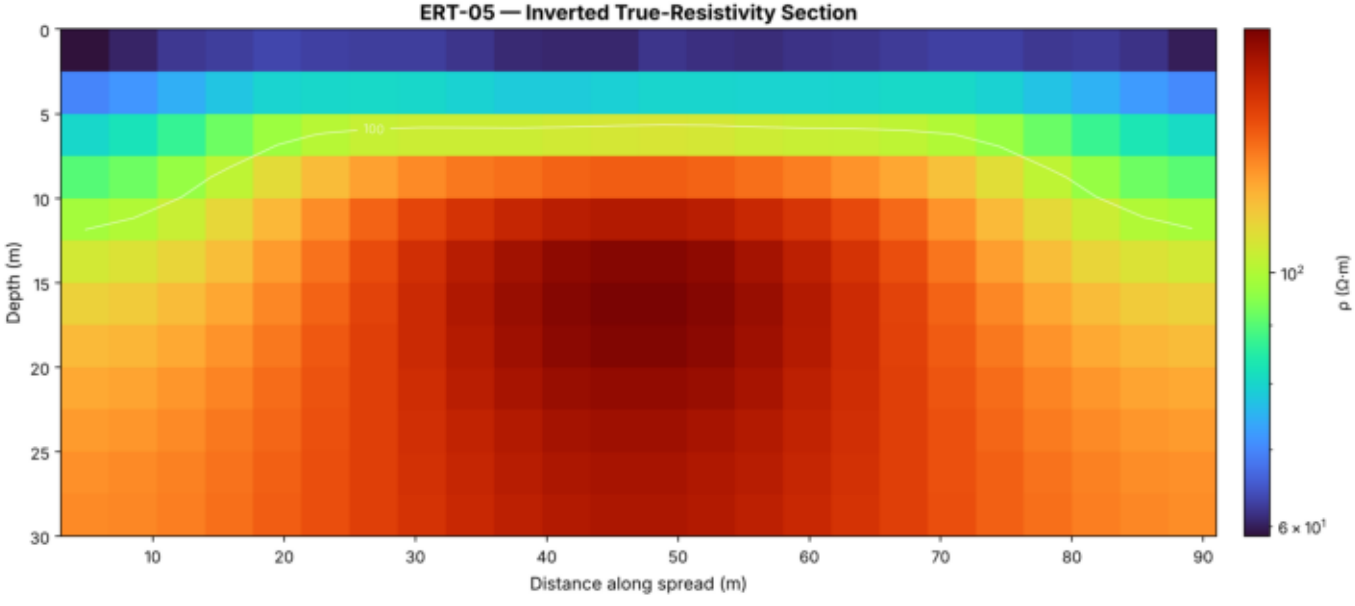
Occam smoothness-constrained least-squares (Loke & Barker 1996).

ERT-05 — Apparent Resistivity Pseudosection (BH5, Group G3)



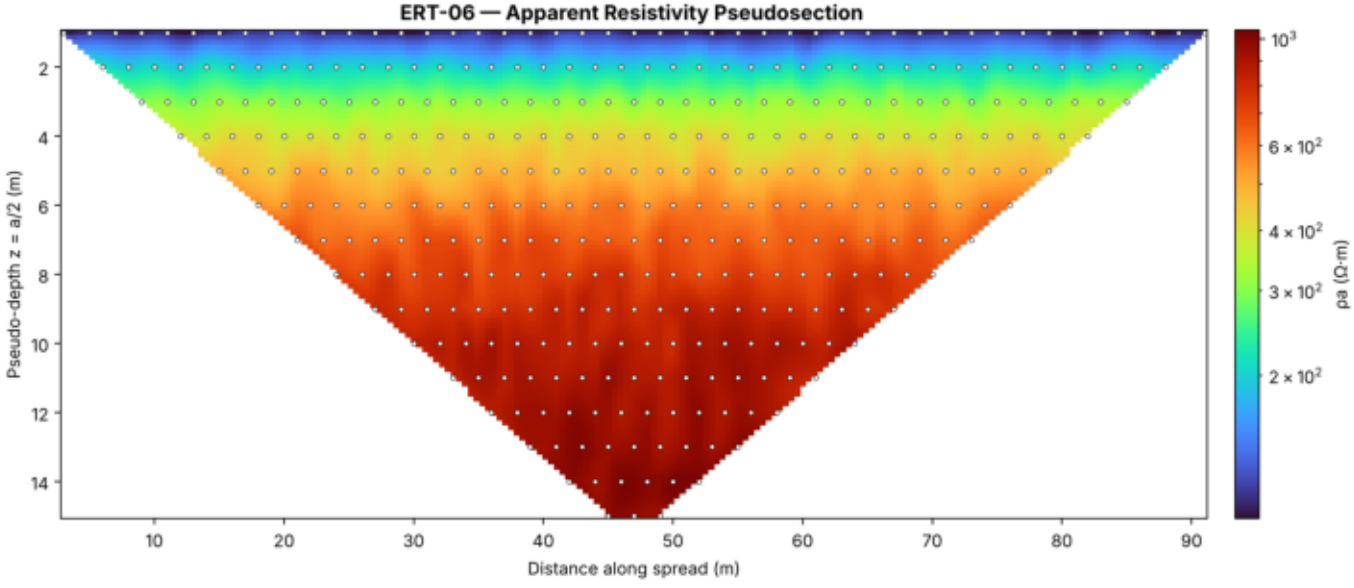
Wenner-alpha, 48 electrodes @ 2 m, spread 100 m, azimuth N67.6°E.

ERT-05 — Inverted True-Resistivity Section (BH5, Group G3)



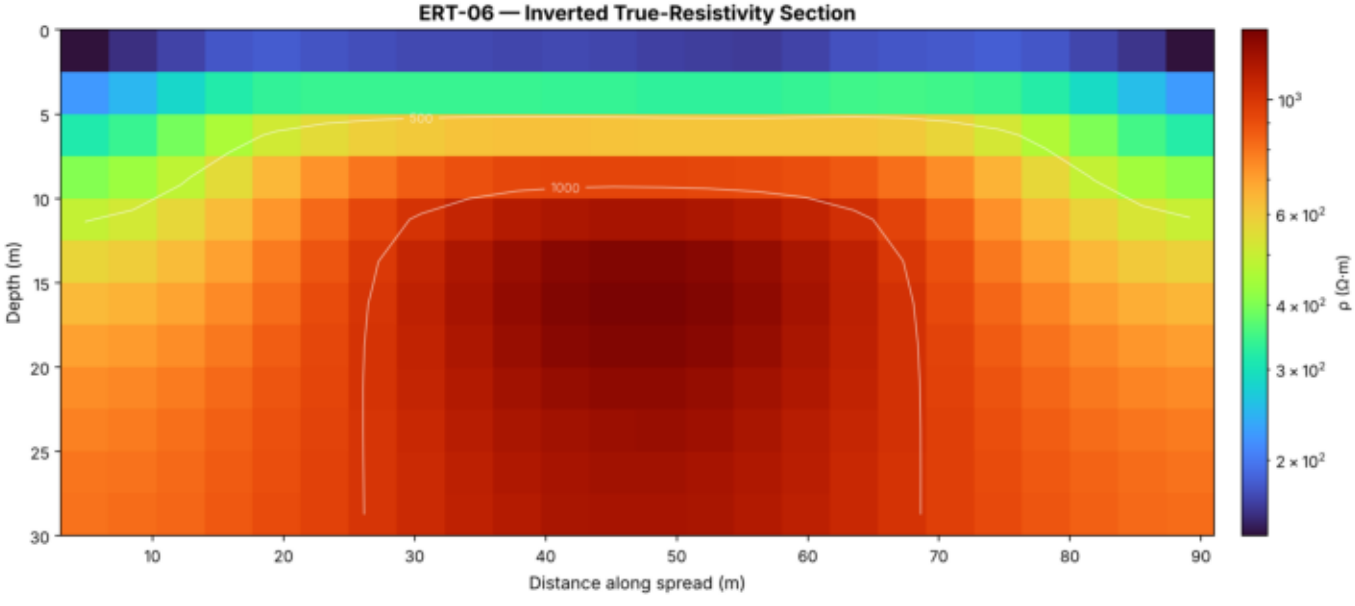
Occam smoothness-constrained least-squares (Loke & Barker 1996).

ERT-06 — Apparent Resistivity Pseudosection (BH6, Group G1)



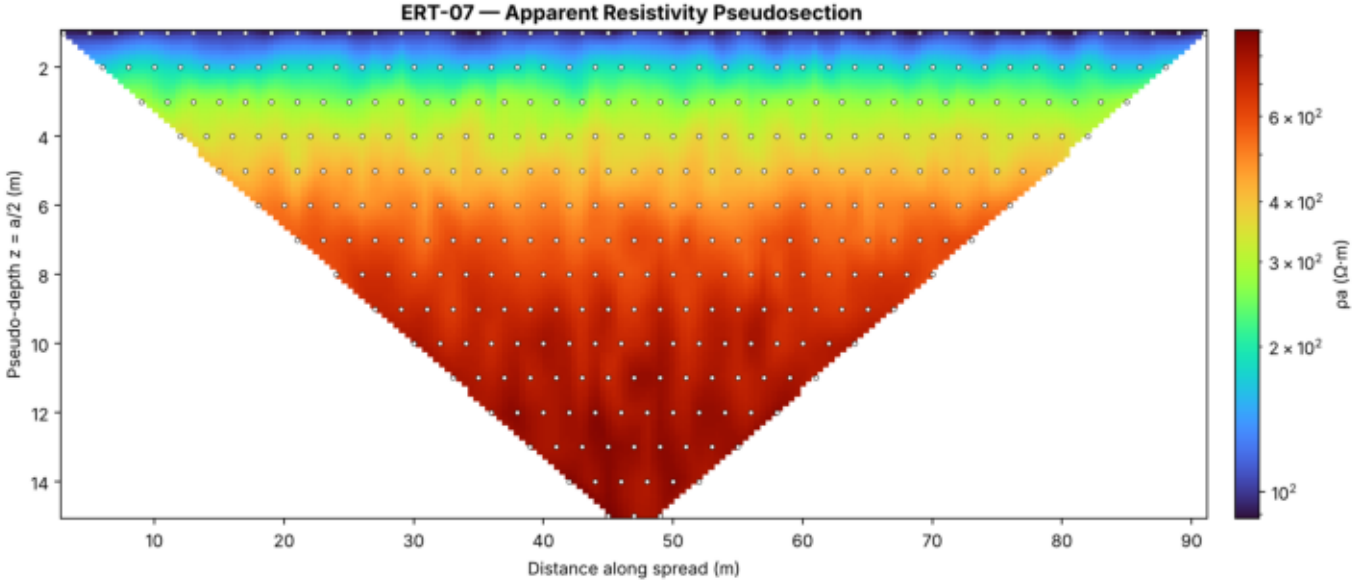
Wenner-alpha, 48 electrodes @ 2 m, spread 100 m, azimuth N67.6°E.

ERT-06 — Inverted True-Resistivity Section (BH6, Group G1)



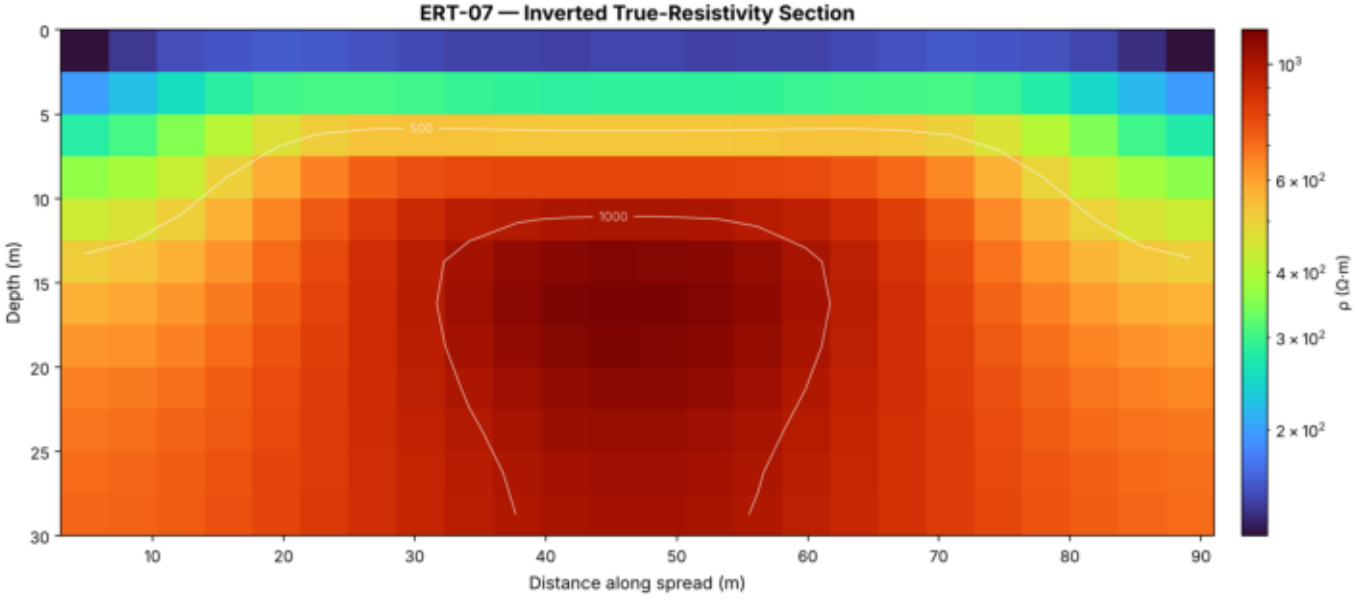
Occam smoothness-constrained least-squares (Loke & Barker 1996).

ERT-07 — Apparent Resistivity Pseudosection (BH7, Group G1)



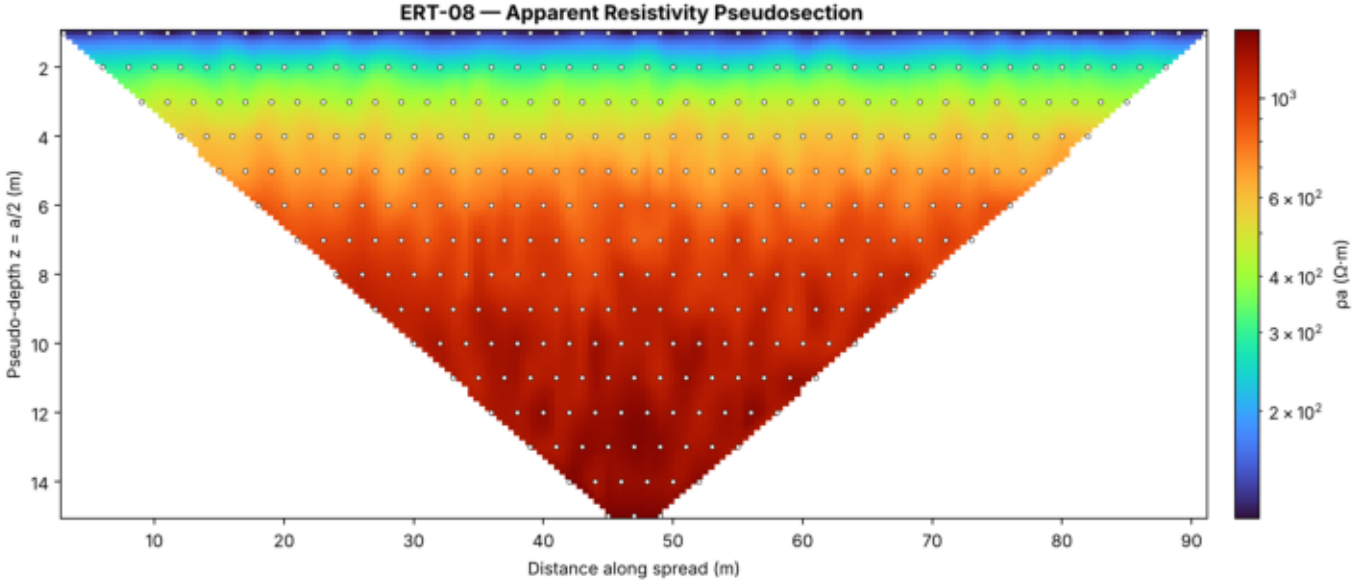
Wenner-alpha, 48 electrodes @ 2 m, spread 100 m, azimuth N67.6°E.

ERT-07 — Inverted True-Resistivity Section (BH7, Group G1)



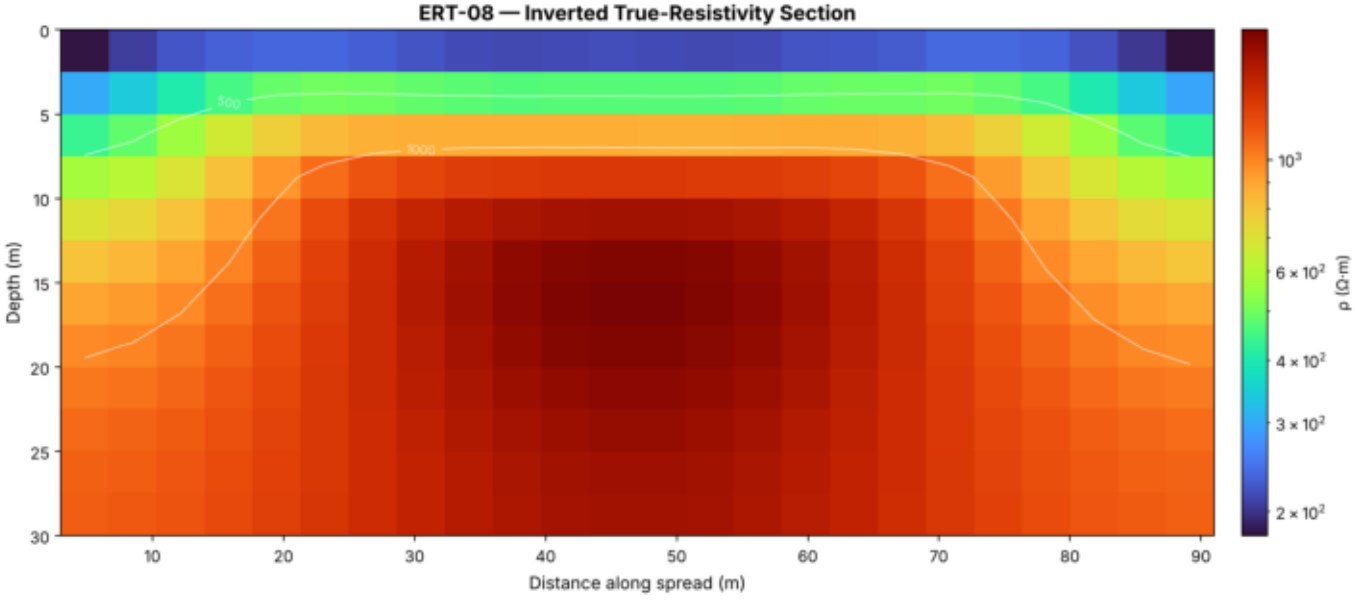
Occam smoothness-constrained least-squares (Loke & Barker 1996).

ERT-08 — Apparent Resistivity Pseudosection (BH8, Group G1)



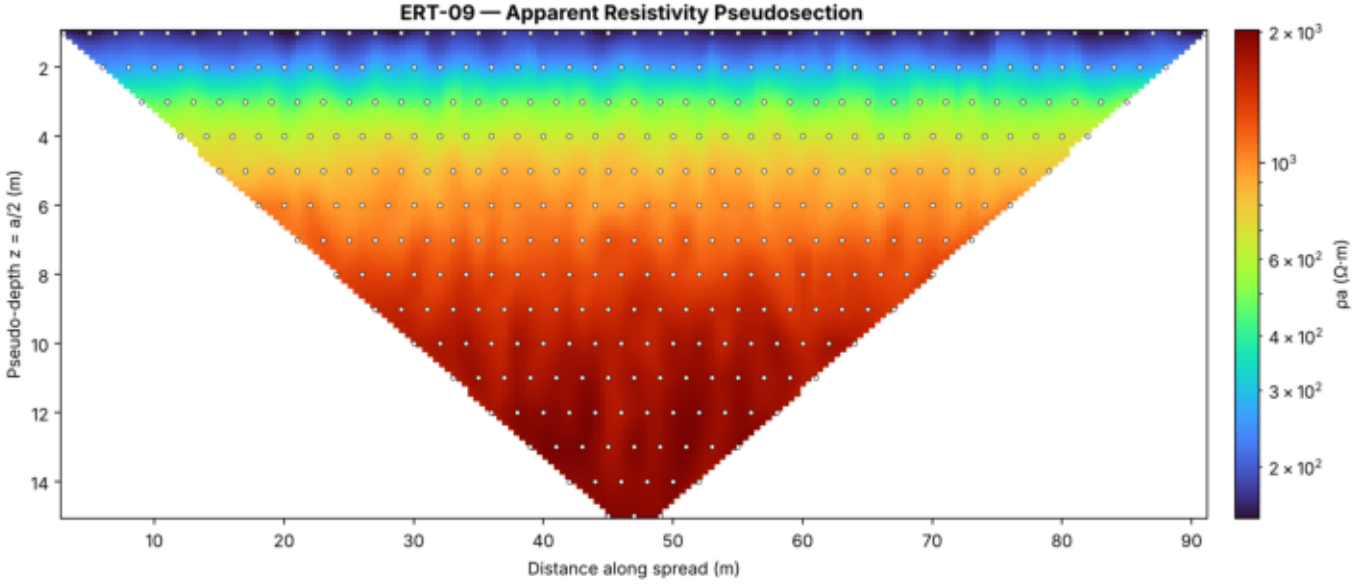
Wenner-alpha, 48 electrodes @ 2 m, spread 100 m, azimuth N67.6°E.

ERT-08 — Inverted True-Resistivity Section (BH8, Group G1)



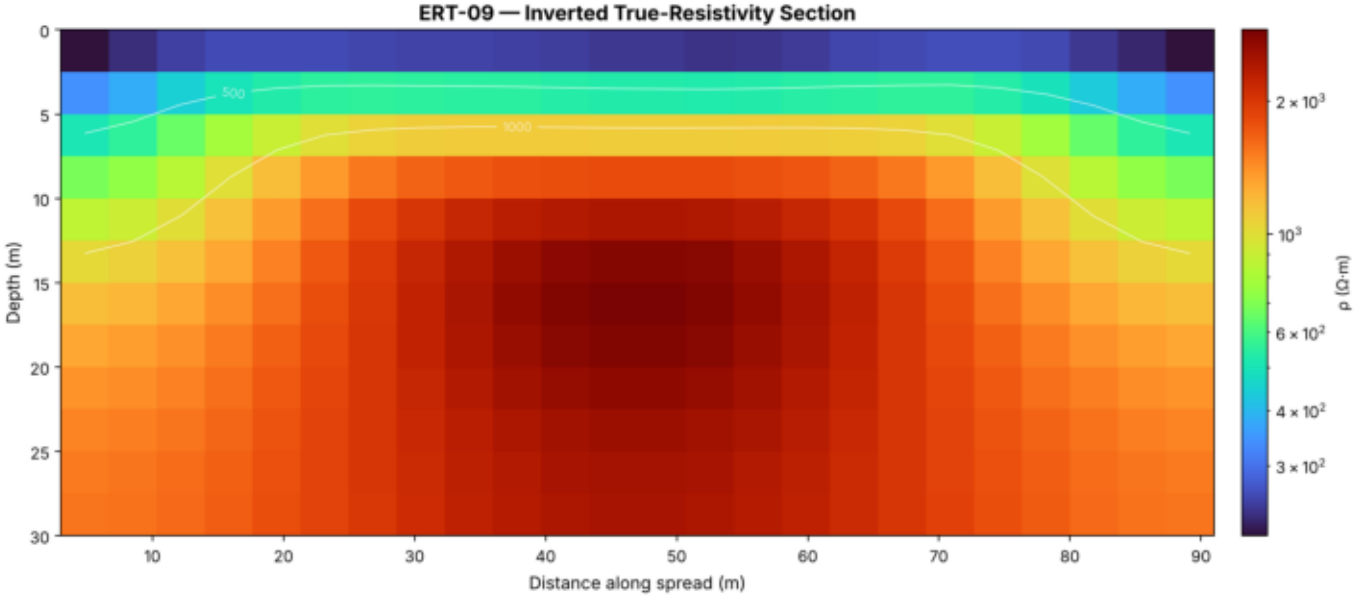
Occam smoothness-constrained least-squares (Loke & Barker 1996).

ERT-09 — Apparent Resistivity Pseudosection (BH9, Group G2)



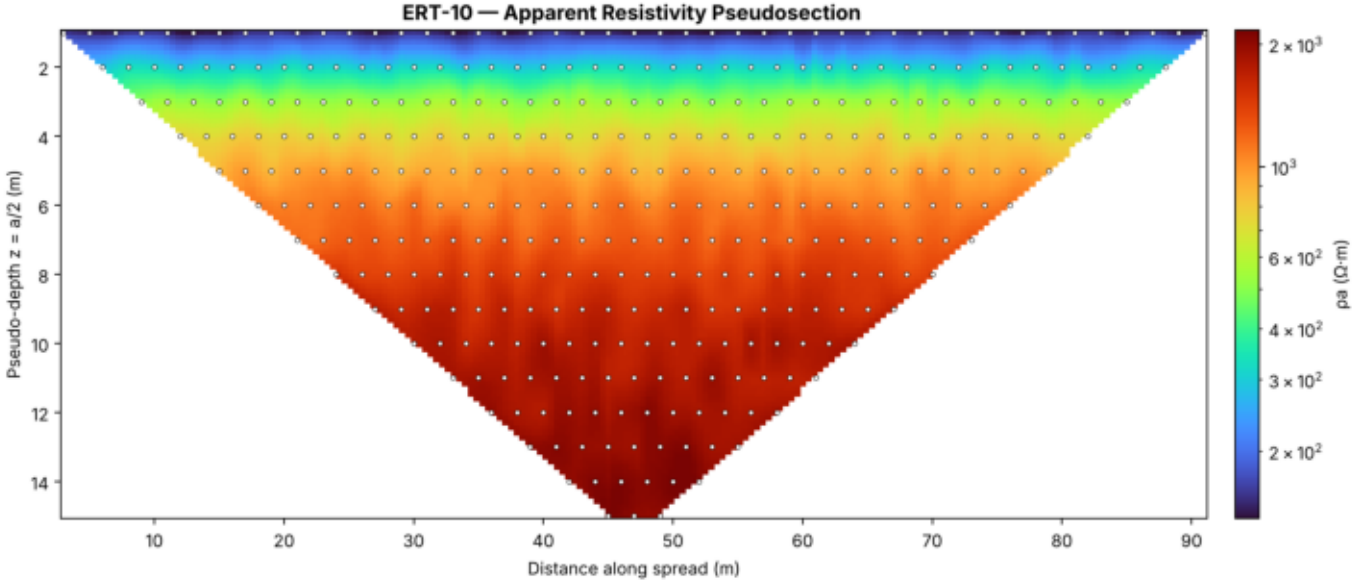
Wenner-alpha, 48 electrodes @ 2 m, spread 100 m, azimuth N67.6°E.

ERT-09 — Inverted True-Resistivity Section (BH9, Group G2)



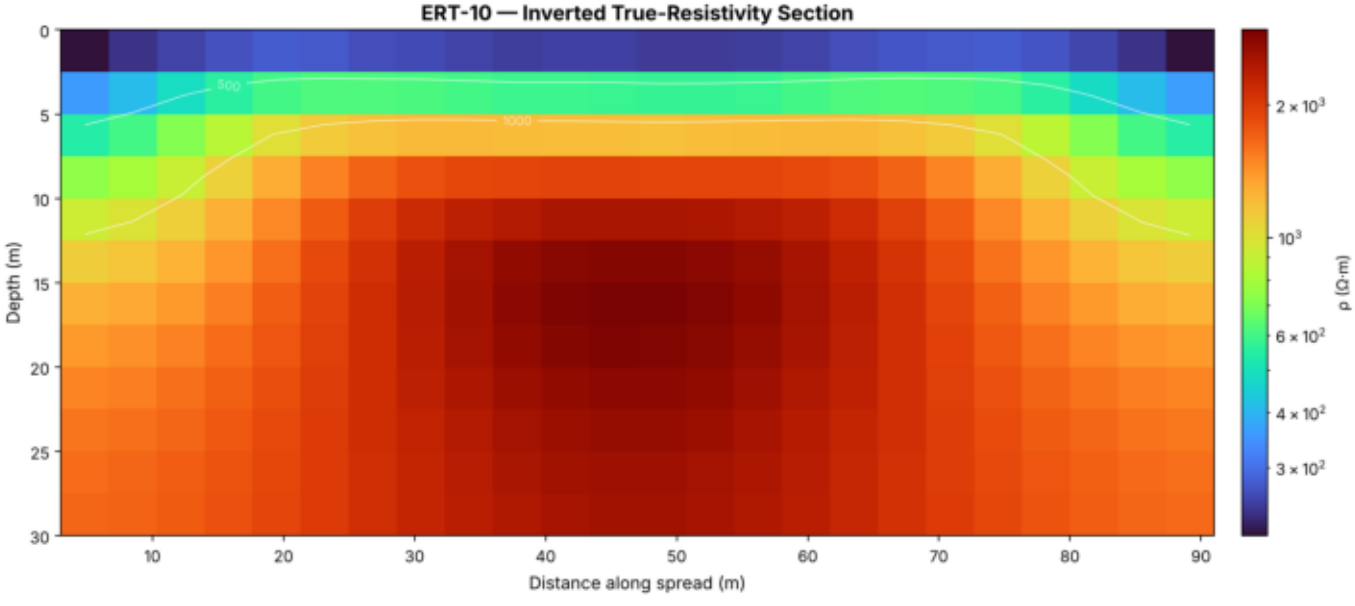
Occam smoothness-constrained least-squares (Loke & Barker 1996).

ERT-10 — Apparent Resistivity Pseudosection (BH10, Group G2)



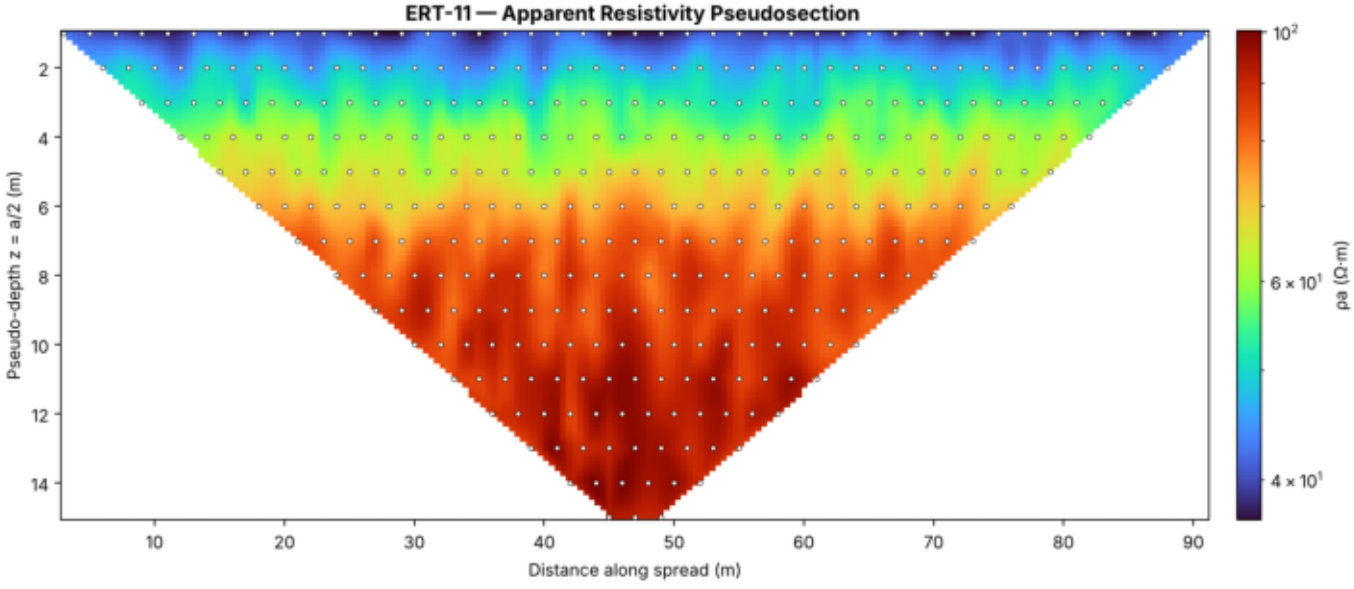
Wenner-alpha, 48 electrodes @ 2 m, spread 100 m, azimuth N67.6°E.

ERT-10 — Inverted True-Resistivity Section (BH10, Group G2)



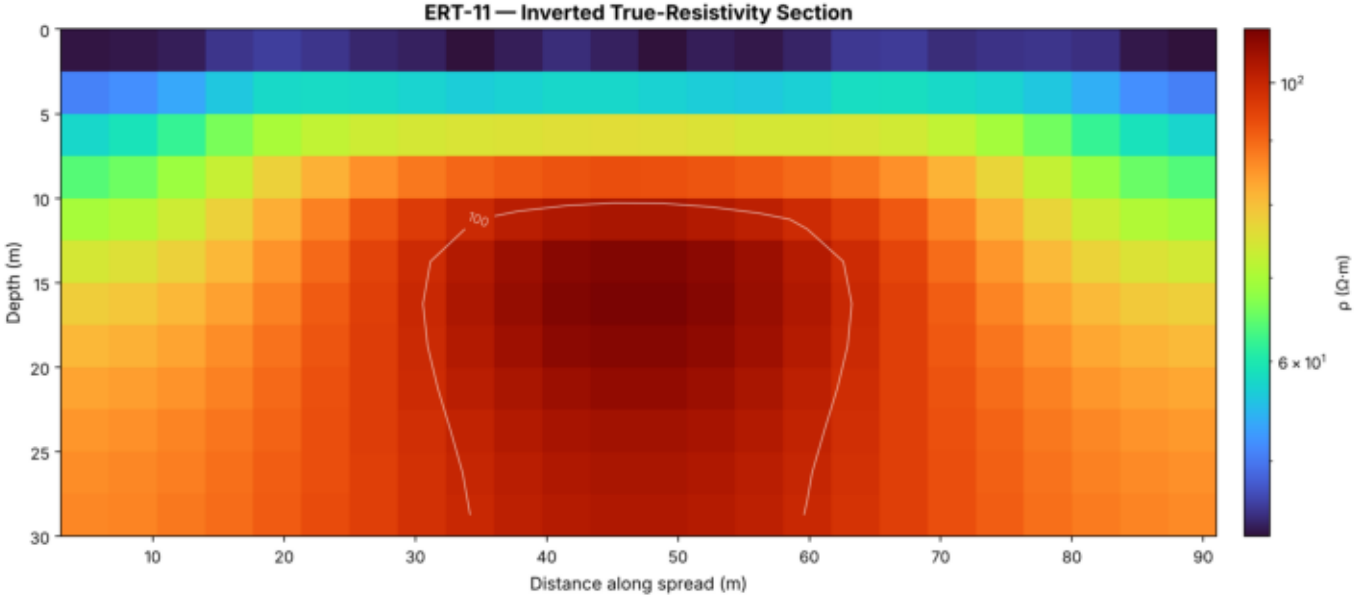
Occam smoothness-constrained least-squares (Loke & Barker 1996).

ERT-11 — Apparent Resistivity Pseudosection (BH11, Group G5)



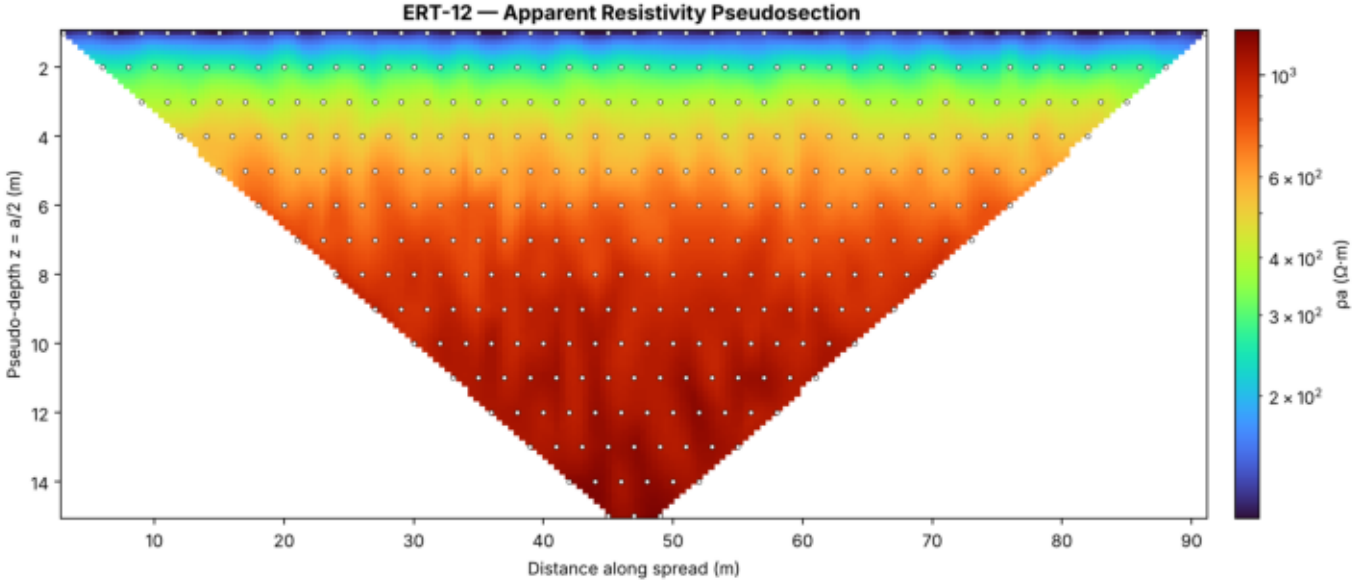
Wenner-alpha, 48 electrodes @ 2 m, spread 100 m, azimuth N67.6°E.

ERT-11 — Inverted True-Resistivity Section (BH11, Group G5)



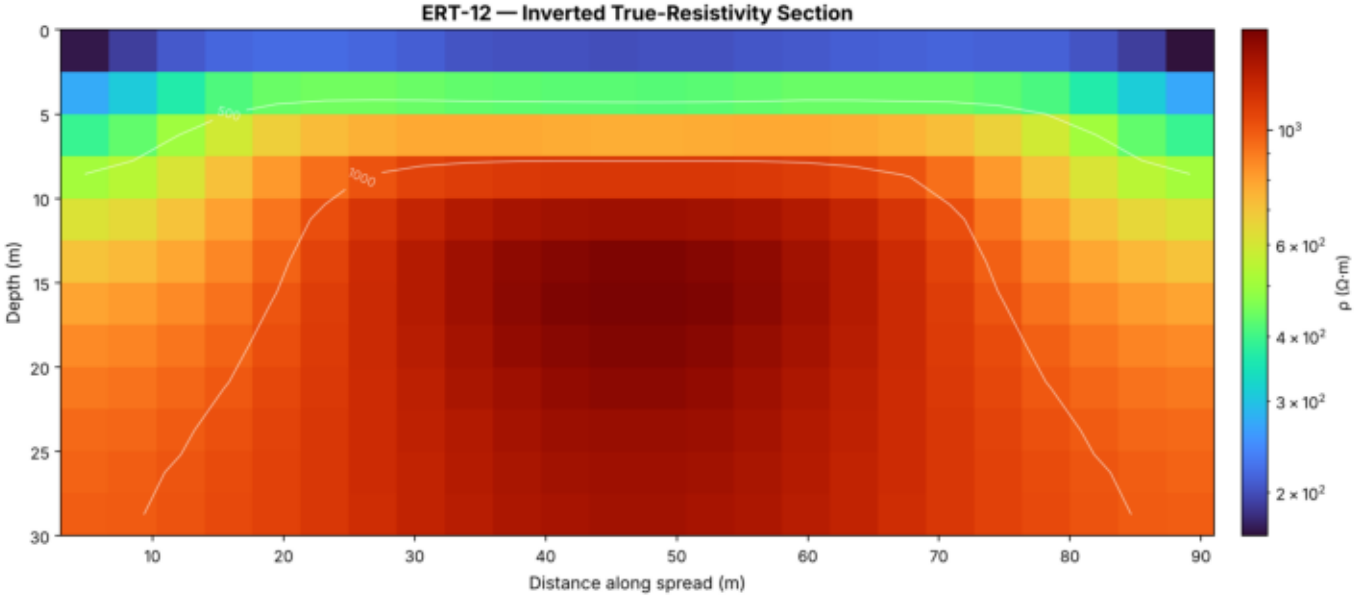
Occam smoothness-constrained least-squares (Loke & Barker 1996).

ERT-12 — Apparent Resistivity Pseudosection (BH12, Group G1)



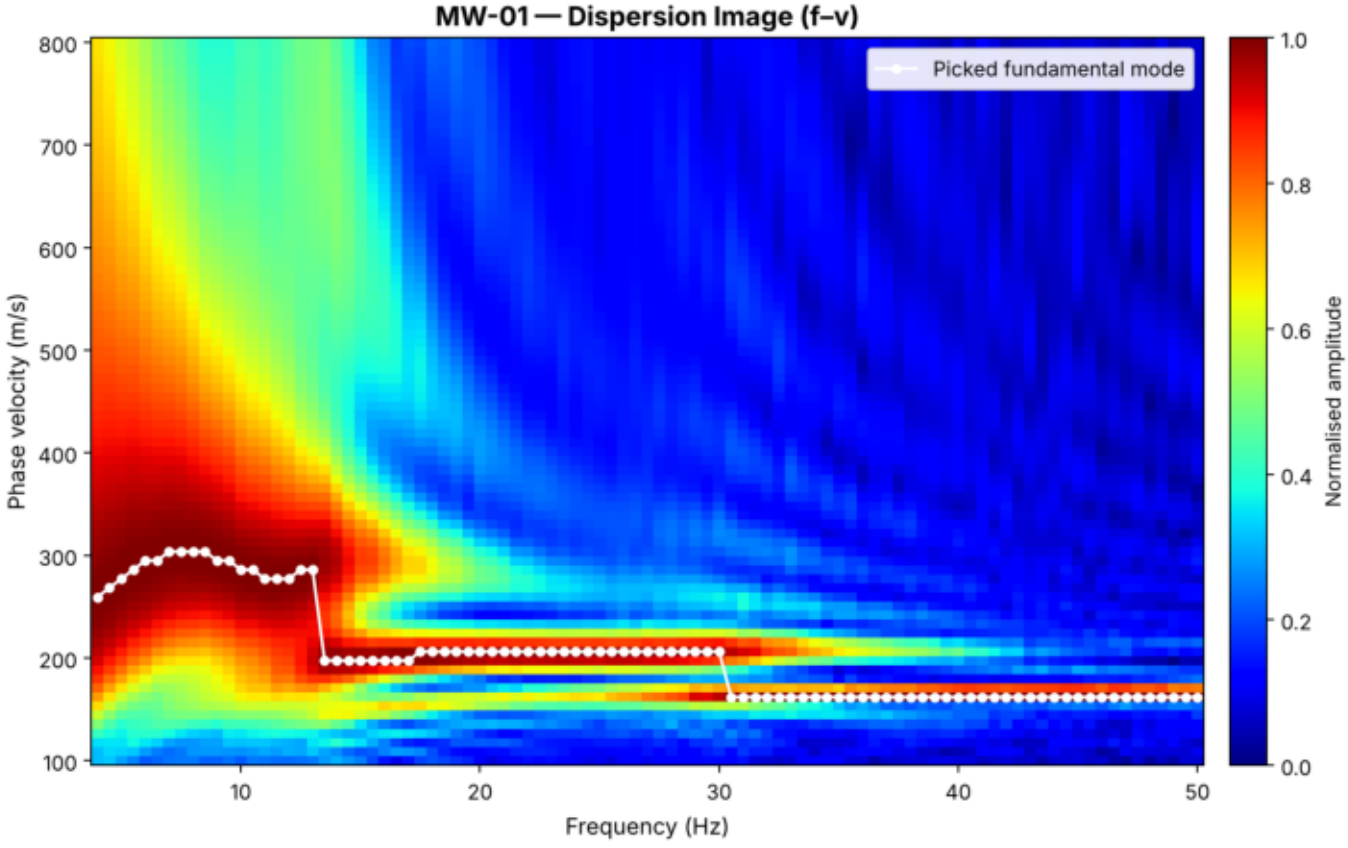
Wenner-alpha, 48 electrodes @ 2 m, spread 100 m, azimuth N67.6°E.

ERT-12 — Inverted True-Resistivity Section (BH12, Group G1)



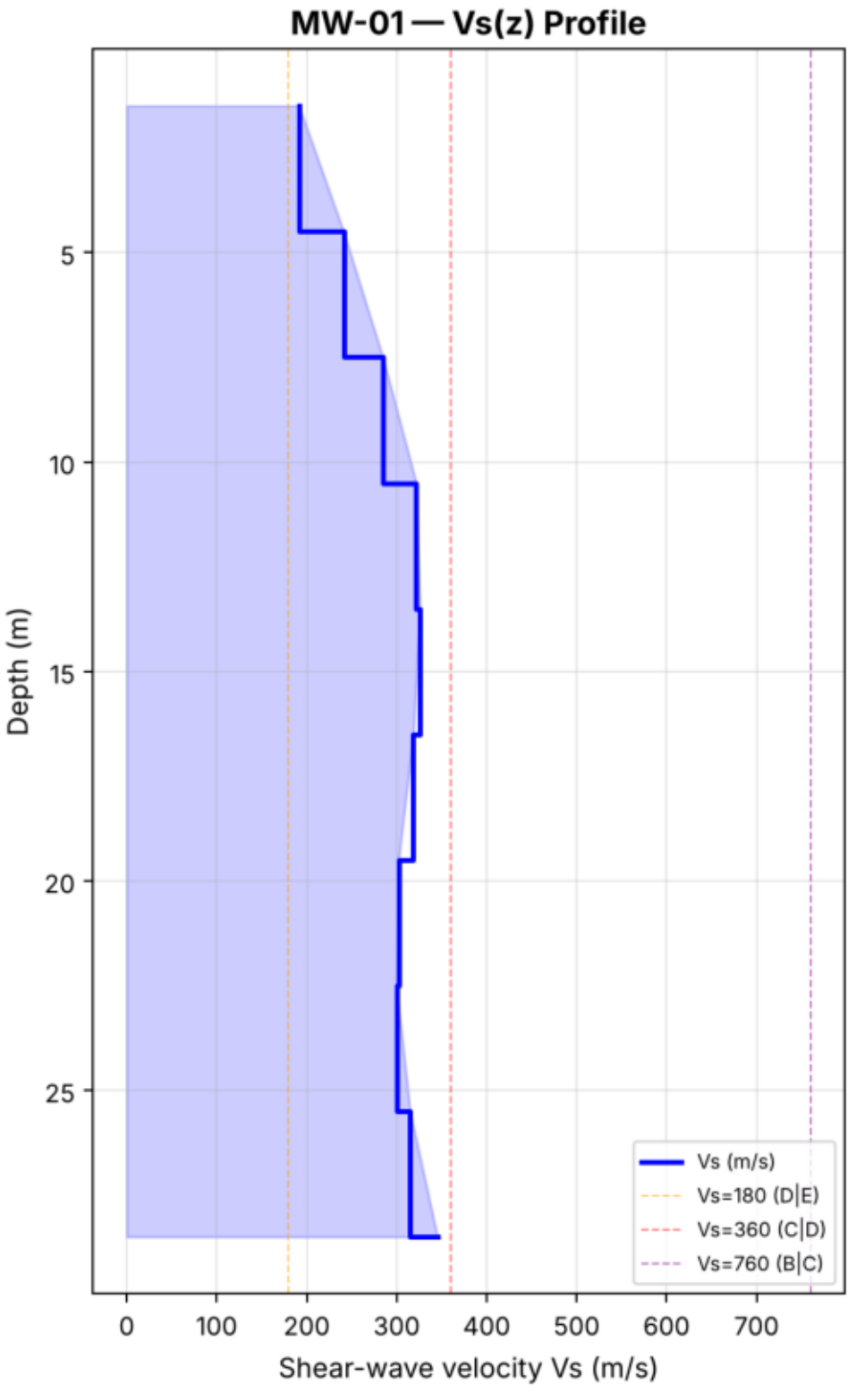
Occam smoothness-constrained least-squares (Loke & Barker 1996).

MW-01 — Dispersion Image (BH1, Group G3)



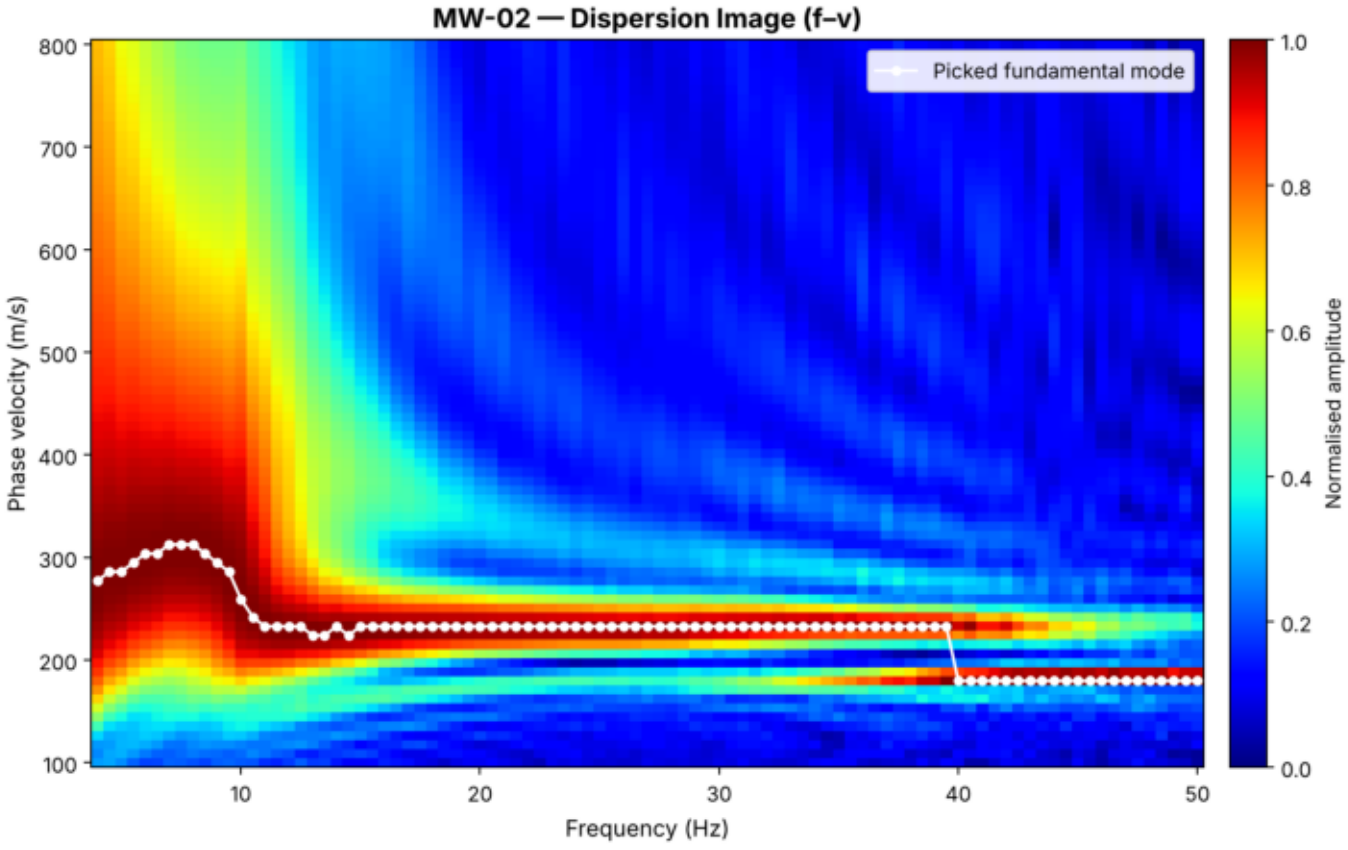
Phase velocity vs frequency, auto-picked fundamental mode.

MW-01 — Vs(z) Profile (BH1, Group G3)



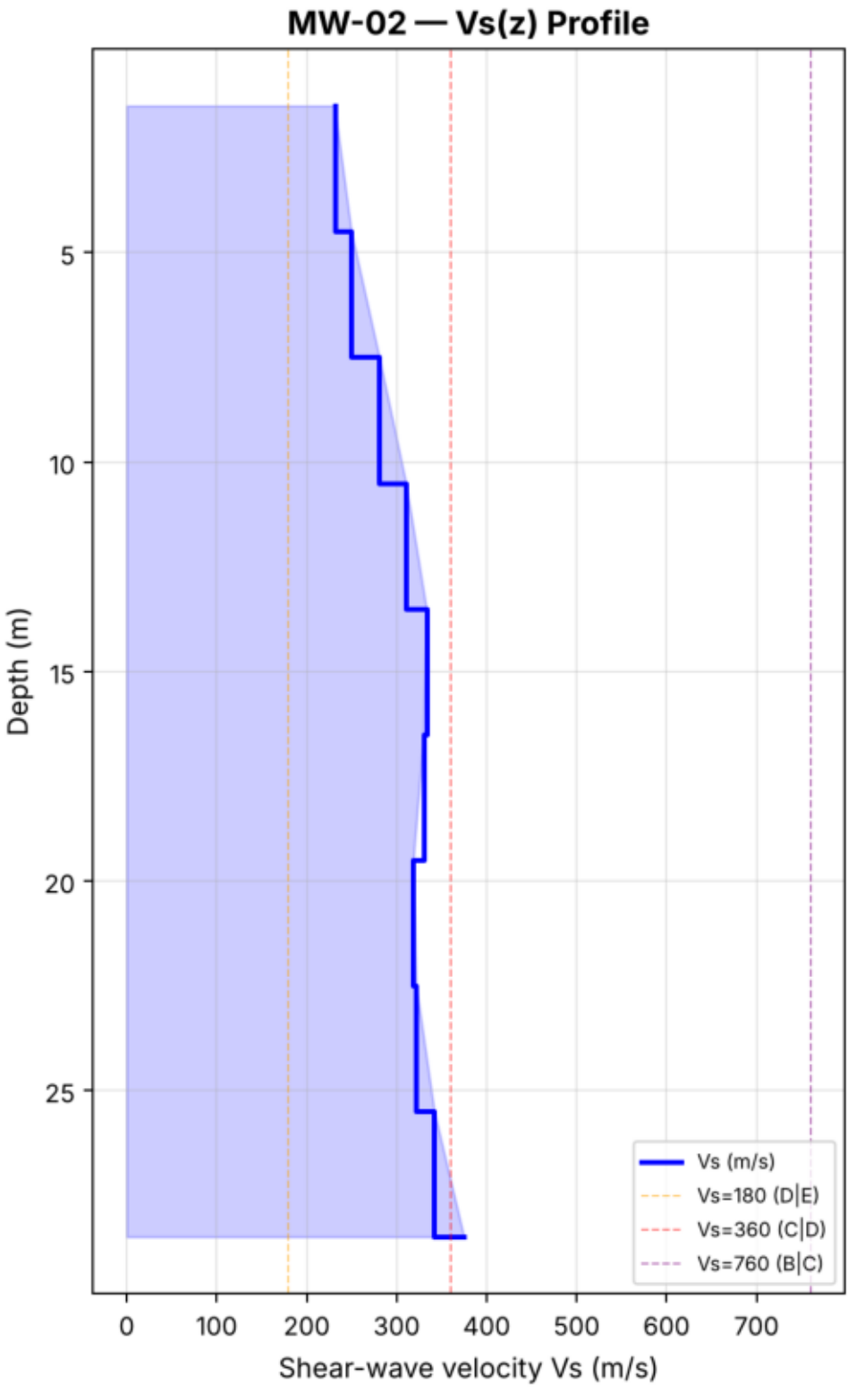
Dispersion inverted to layered Vs profile; a national seismic design code site-class lines overlaid.

MW-02 — Dispersion Image (BH2, Group G3)



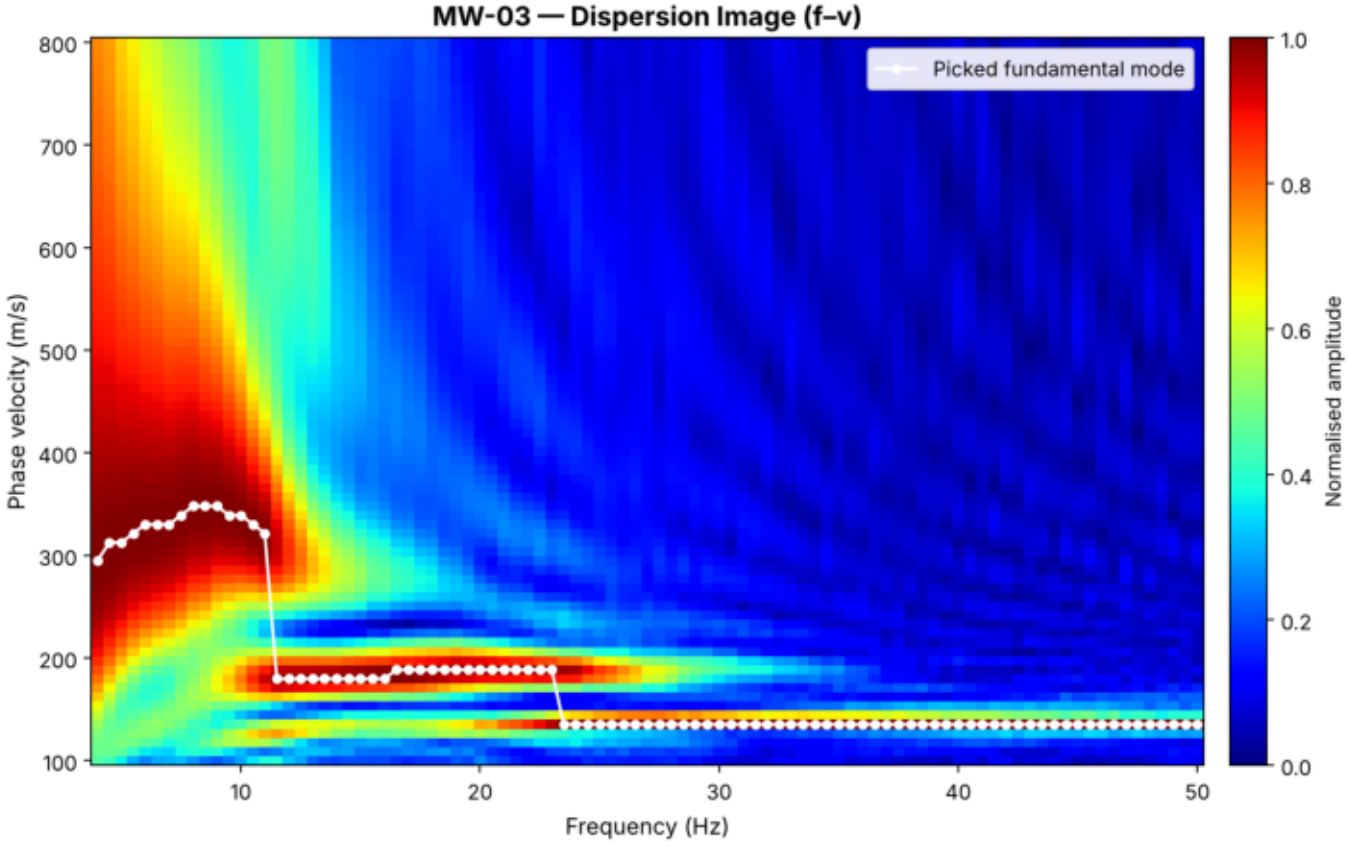
Phase velocity vs frequency, auto-picked fundamental mode.

MW-02 — Vs(z) Profile (BH2, Group G3)



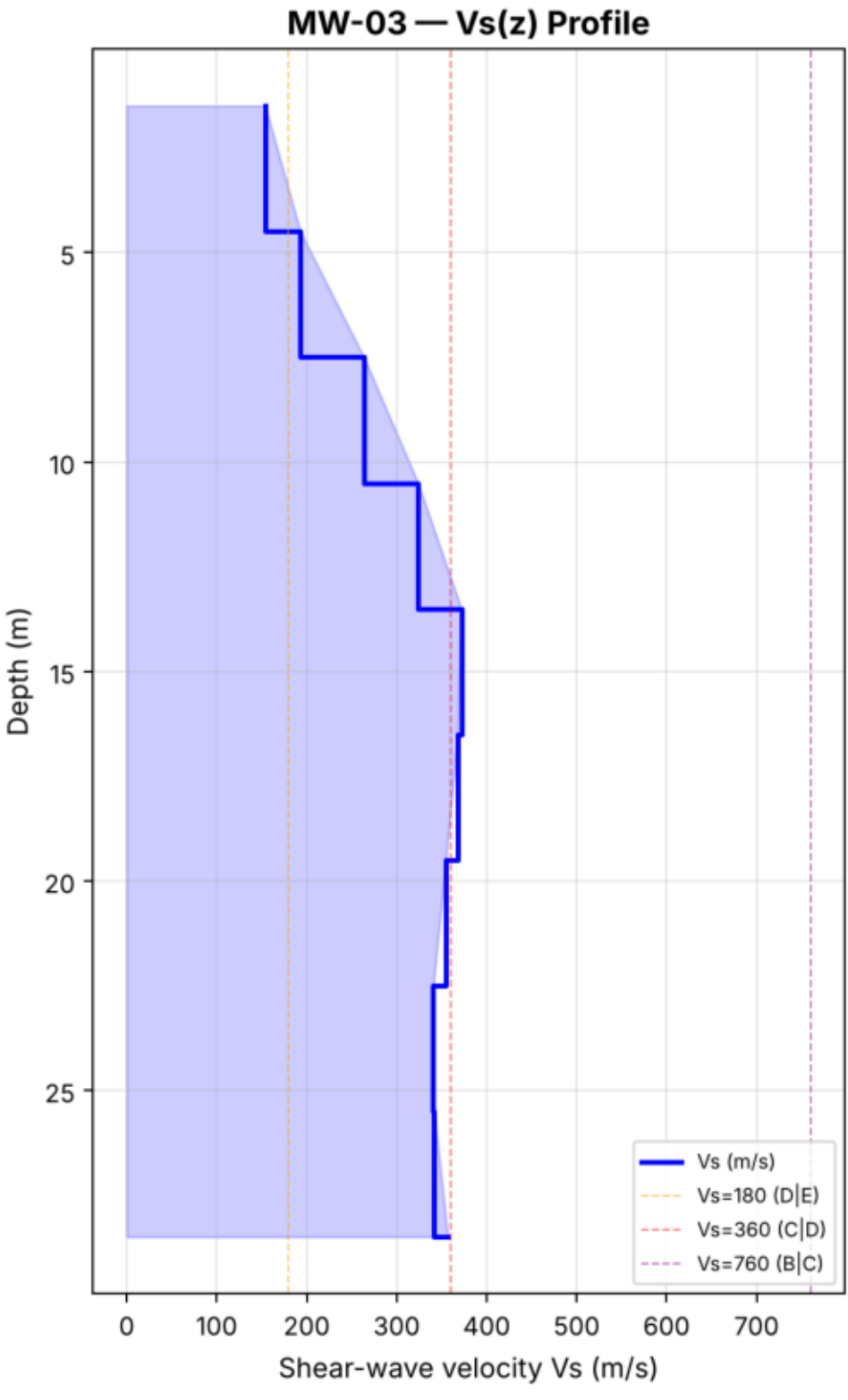
Dispersion inverted to layered Vs profile; a national seismic design code site-class lines overlaid.

MW-03 — Dispersion Image (BH3, Group G5)



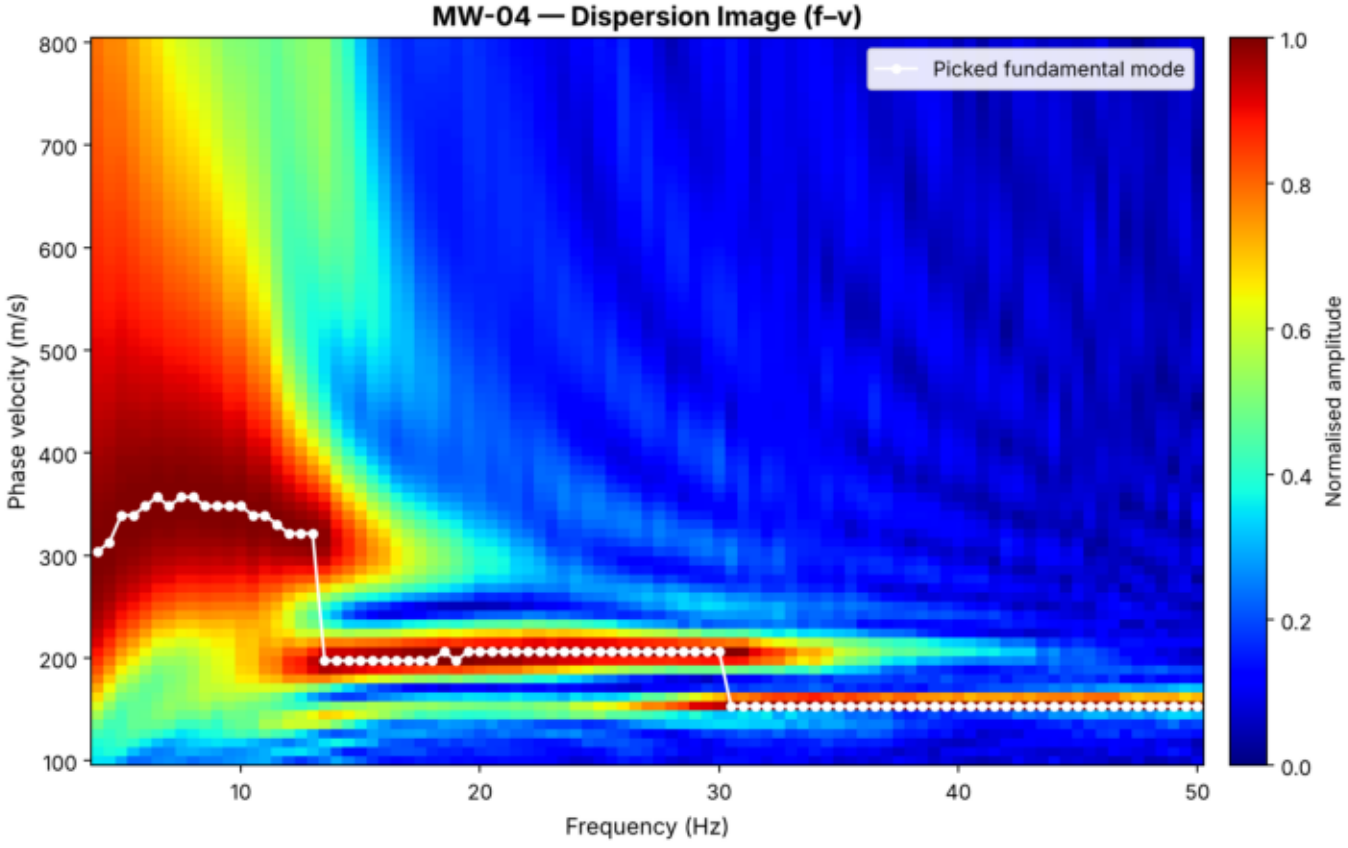
Phase velocity vs frequency, auto-picked fundamental mode.

MW-03 — Vs(z) Profile (BH3, Group G5)



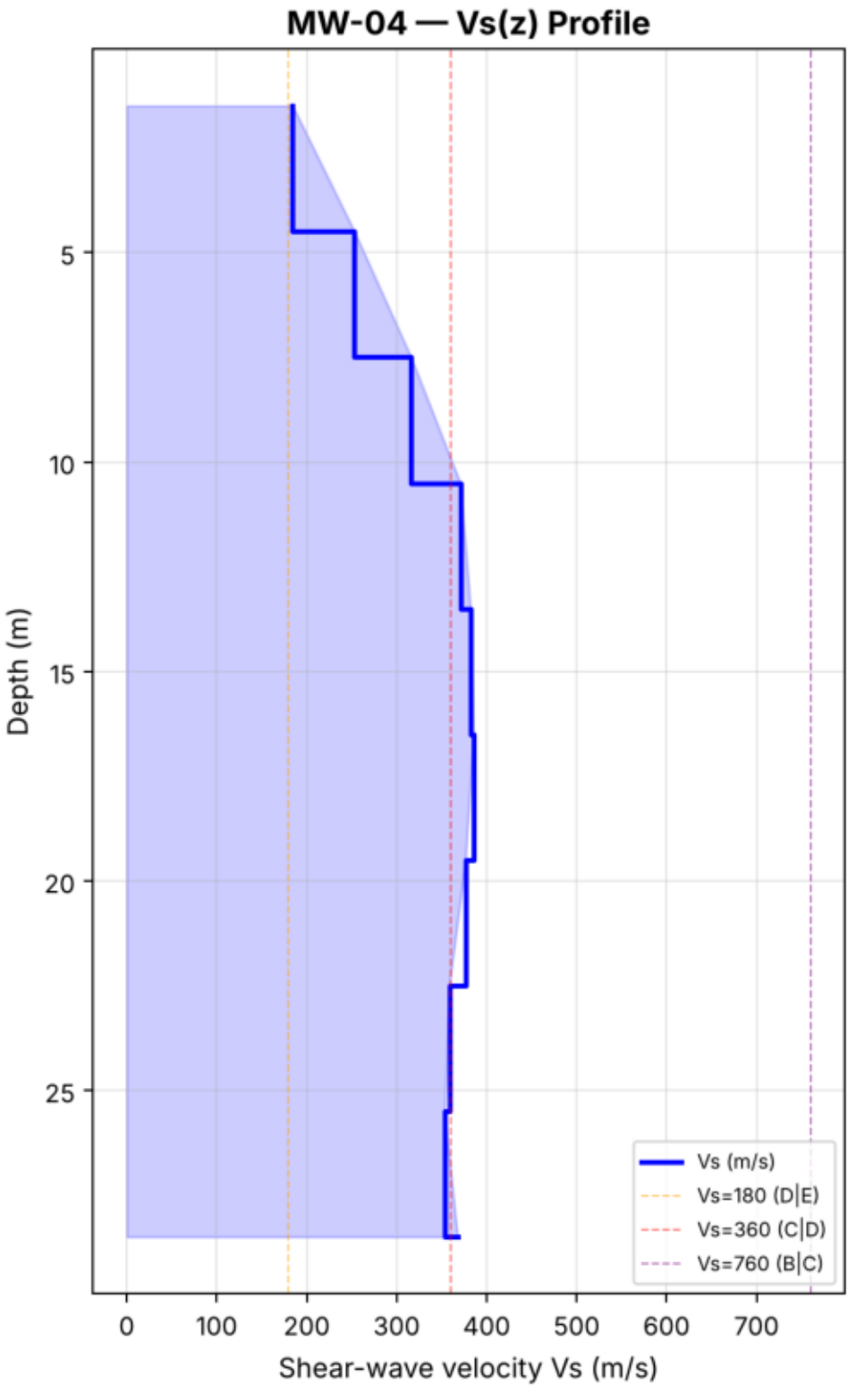
Dispersion inverted to layered Vs profile; a national seismic design code site-class lines overlaid.

MW-04 — Dispersion Image (BH4, Group G5)



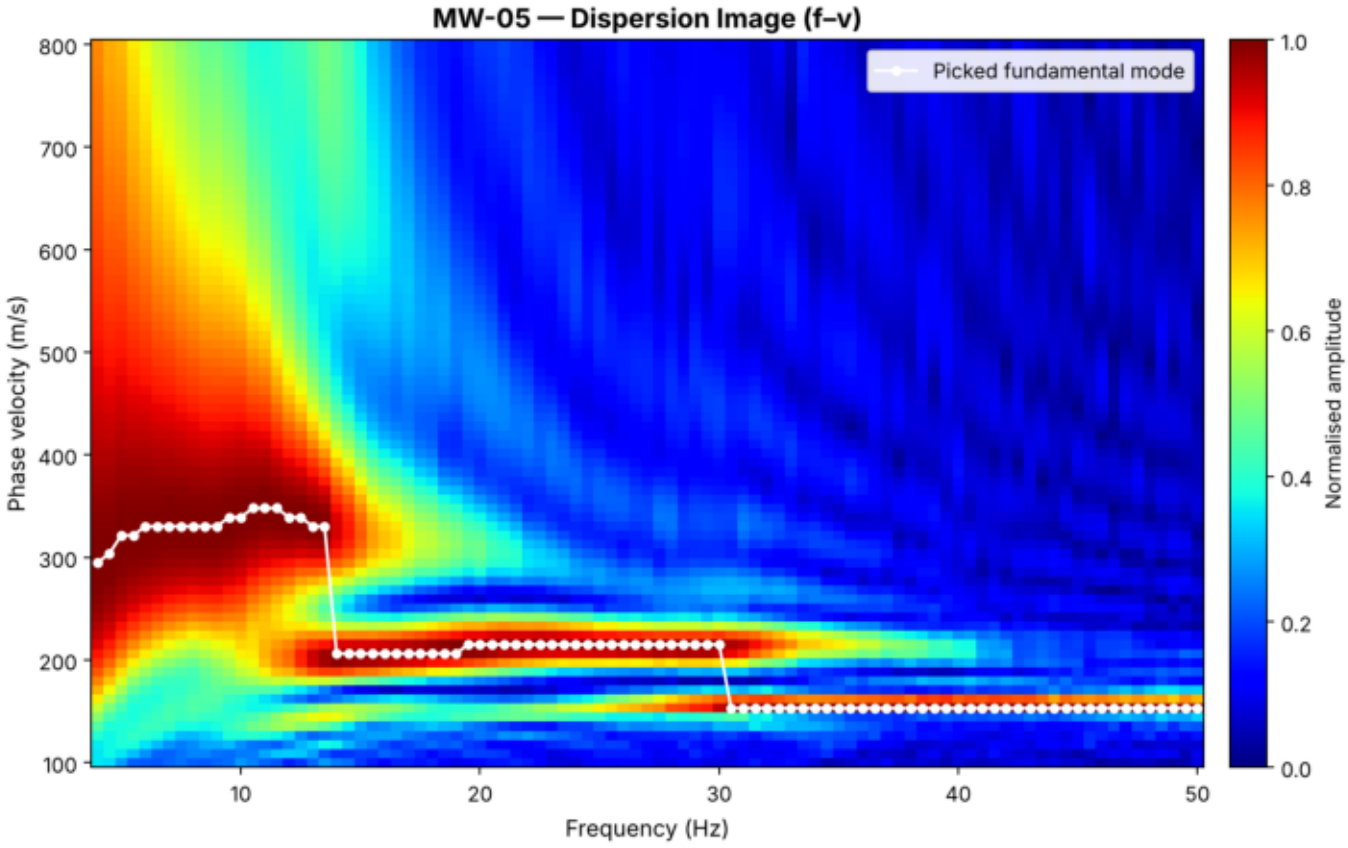
Phase velocity vs frequency, auto-picked fundamental mode.

MW-04 — Vs(z) Profile (BH4, Group G5)



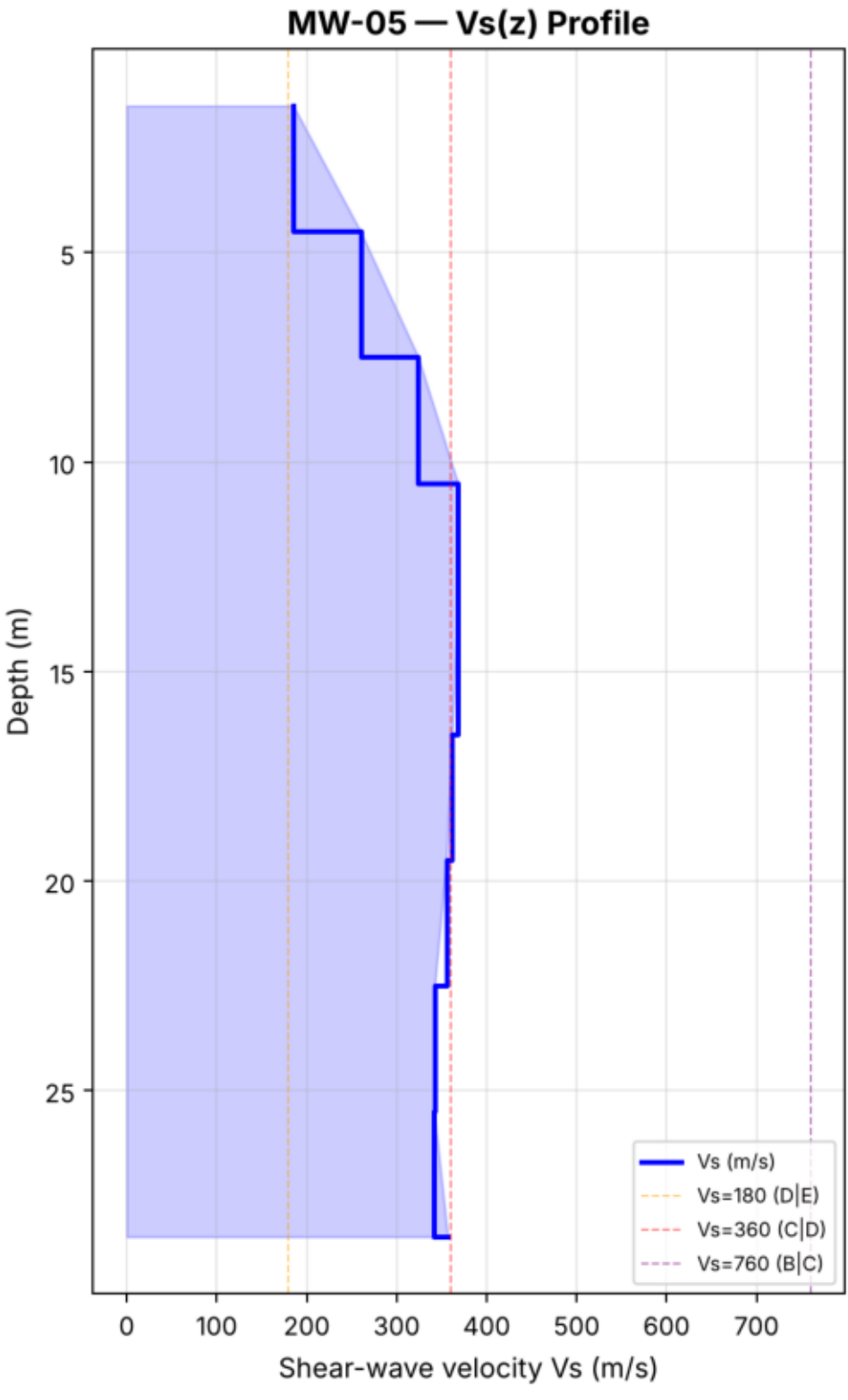
Dispersion inverted to layered Vs profile; a national seismic design code site-class lines overlaid.

MW-05 — Dispersion Image (BH5, Group G3)



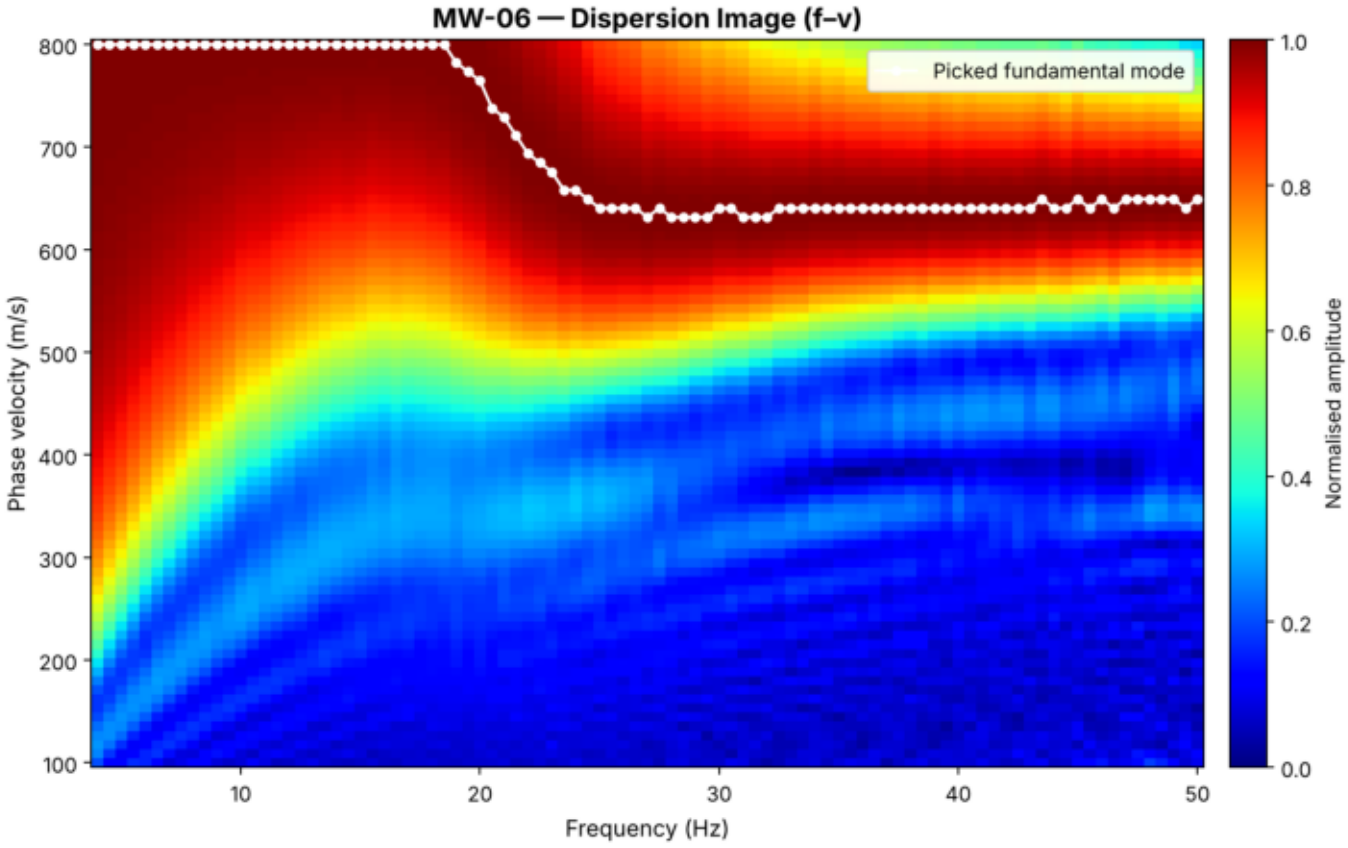
Phase velocity vs frequency, auto-picked fundamental mode.

MW-05 — Vs(z) Profile (BH5, Group G3)



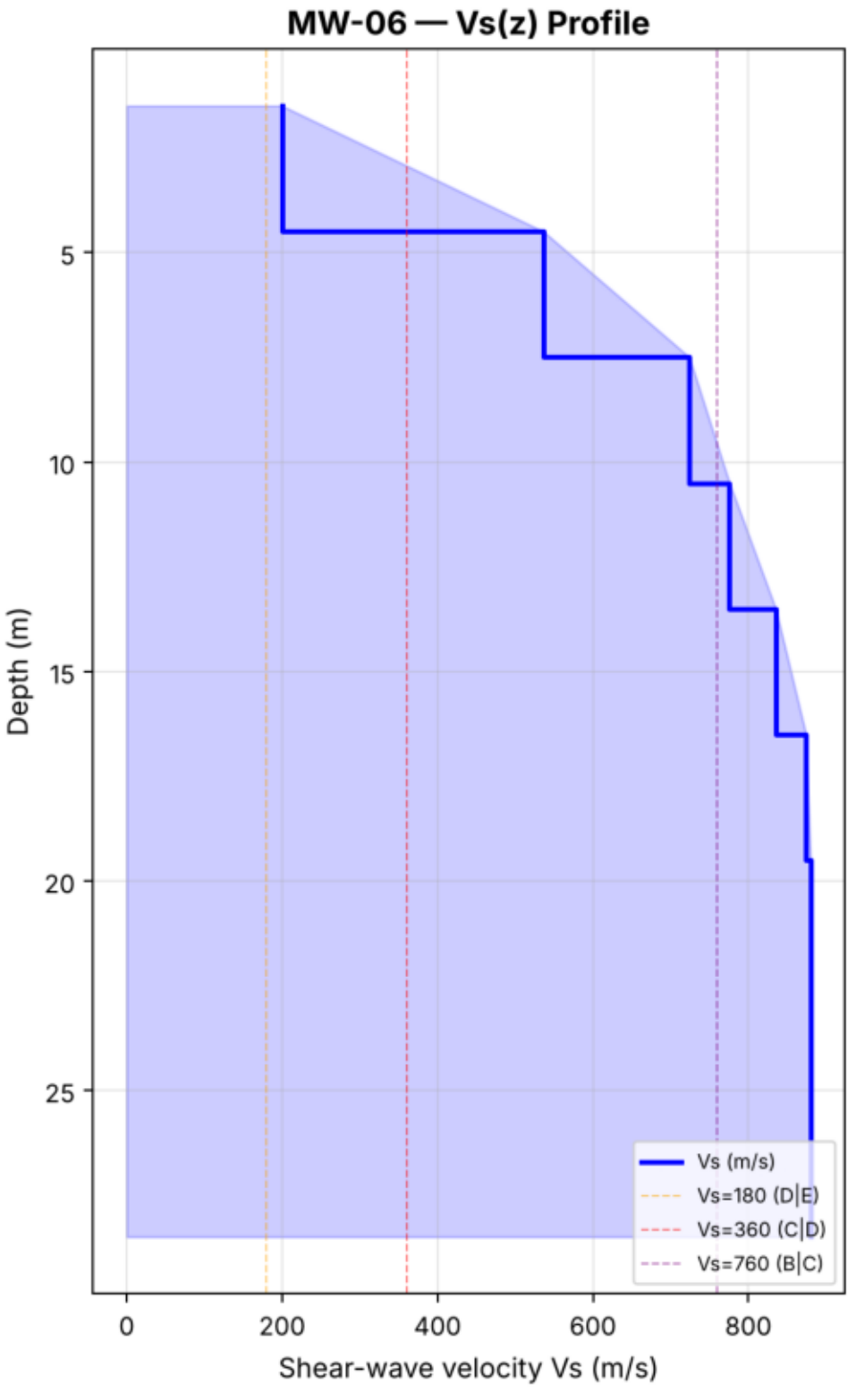
Dispersion inverted to layered Vs profile; a national seismic design code site-class lines overlaid.

MW-06 — Dispersion Image (BH6, Group G1)



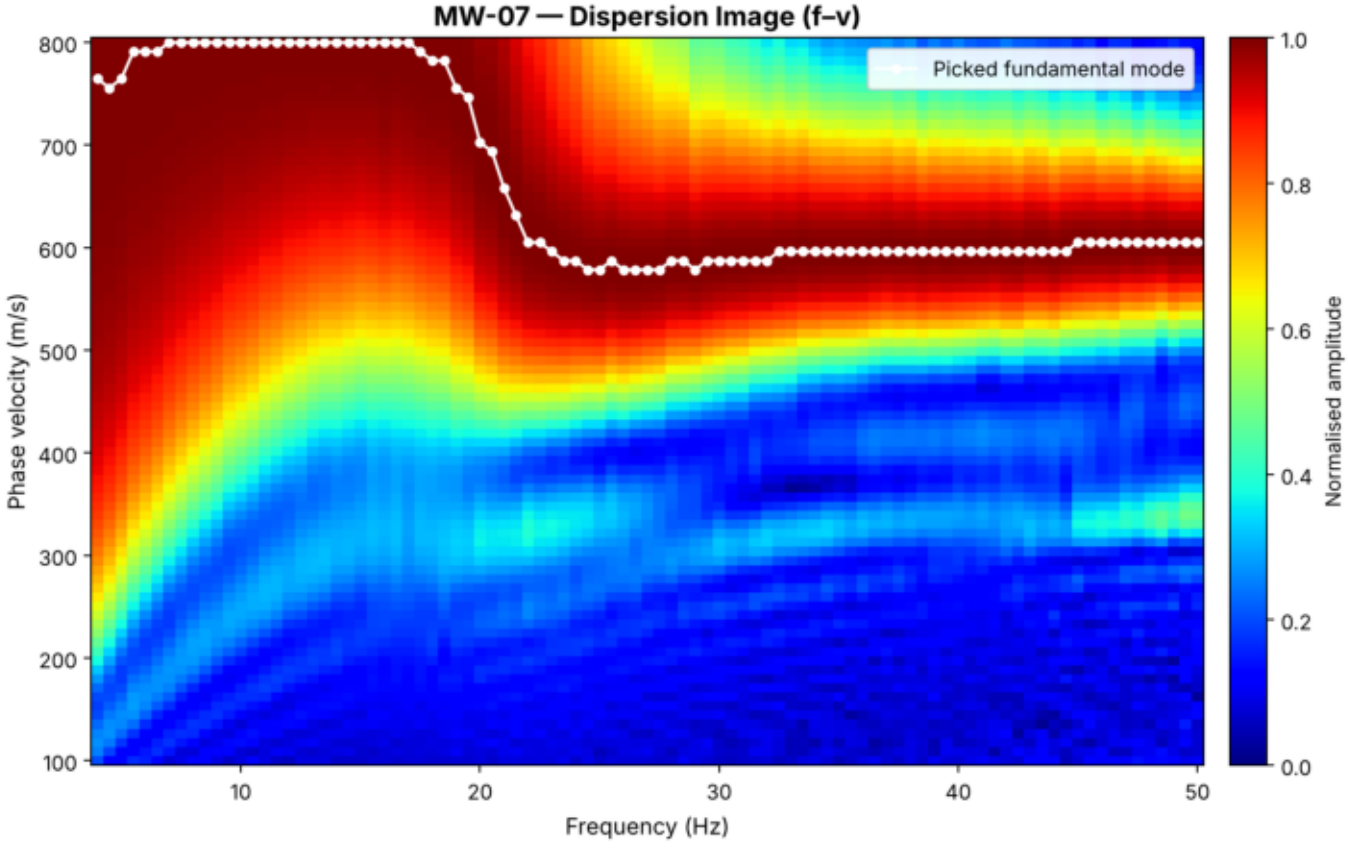
Phase velocity vs frequency, auto-picked fundamental mode.

MW-06 — Vs(z) Profile (BH6, Group G1)



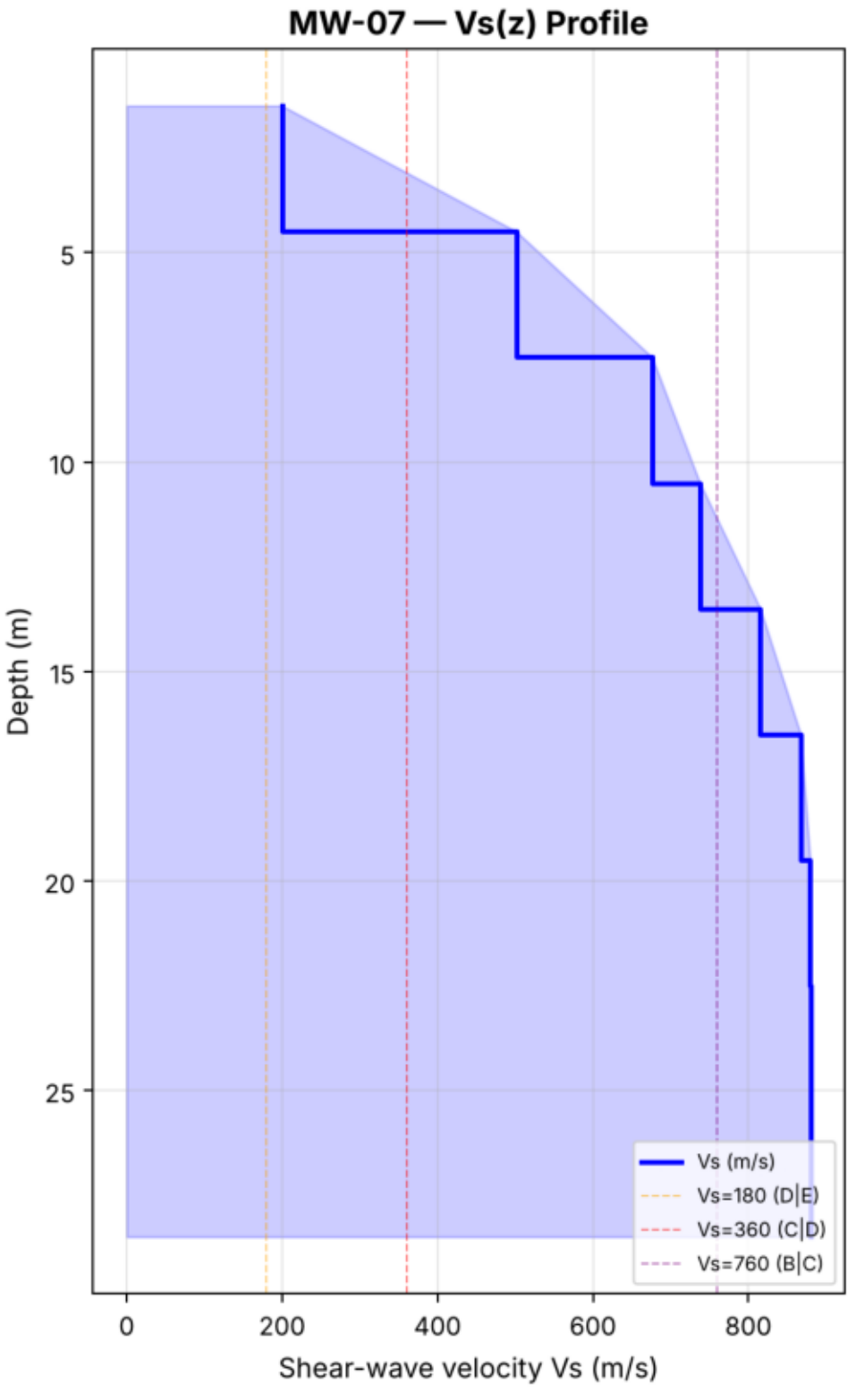
Dispersion inverted to layered Vs profile; a national seismic design code site-class lines overlaid.

MW-07 — Dispersion Image (BH7, Group G1)



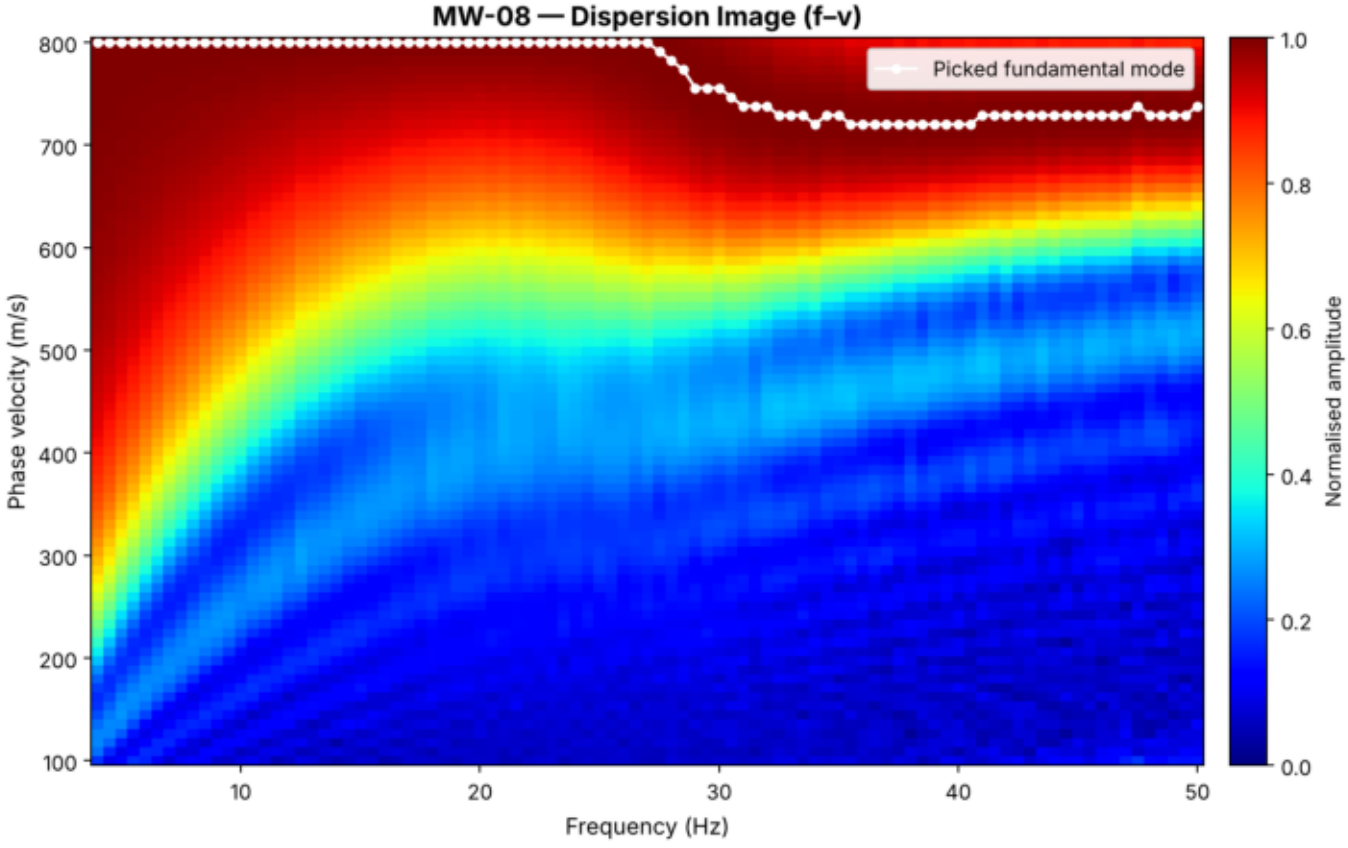
Phase velocity vs frequency, auto-picked fundamental mode.

MW-07 — Vs(z) Profile (BH7, Group G1)



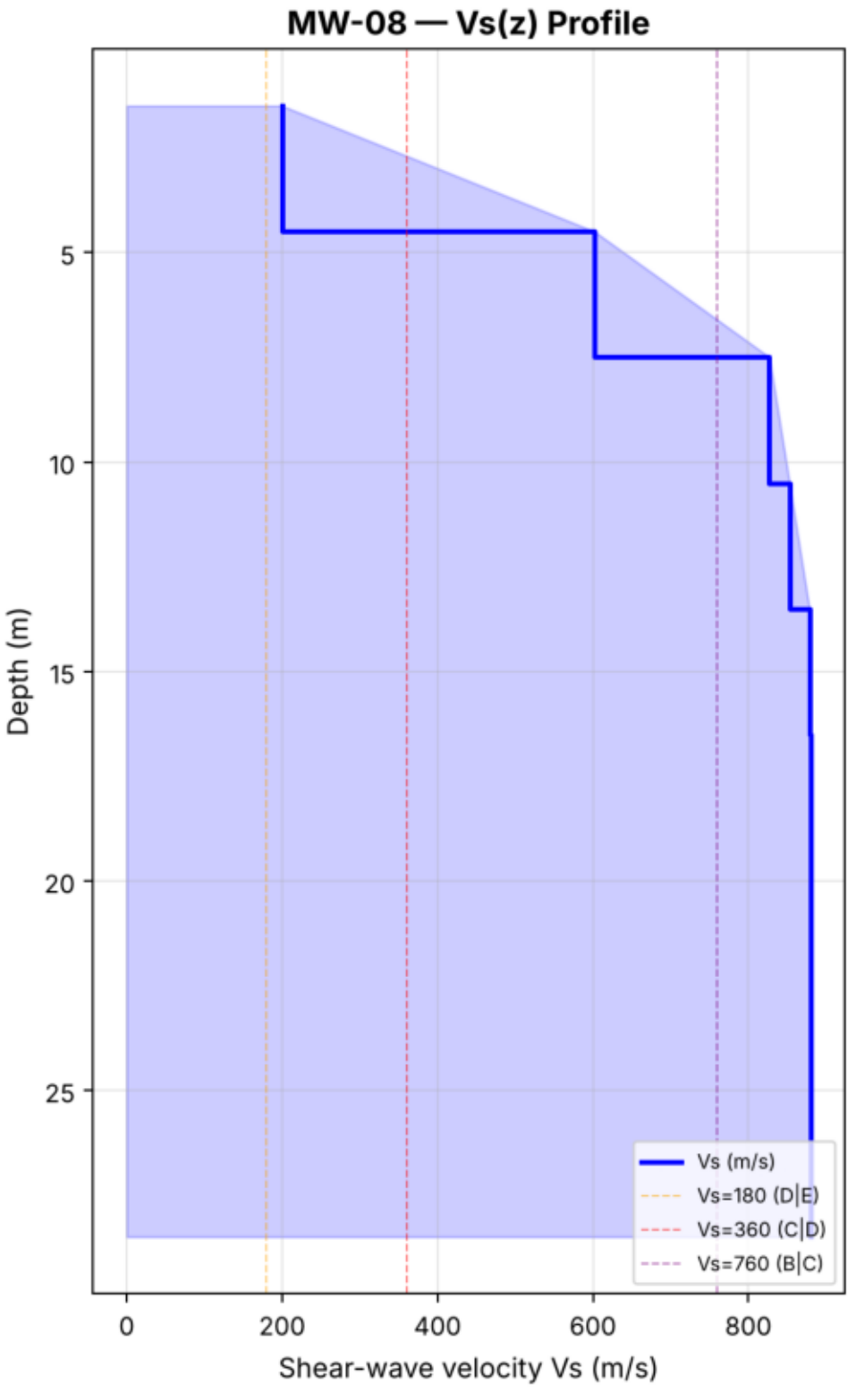
Dispersion inverted to layered Vs profile; a national seismic design code site-class lines overlaid.

MW-08 — Dispersion Image (BH8, Group G1)



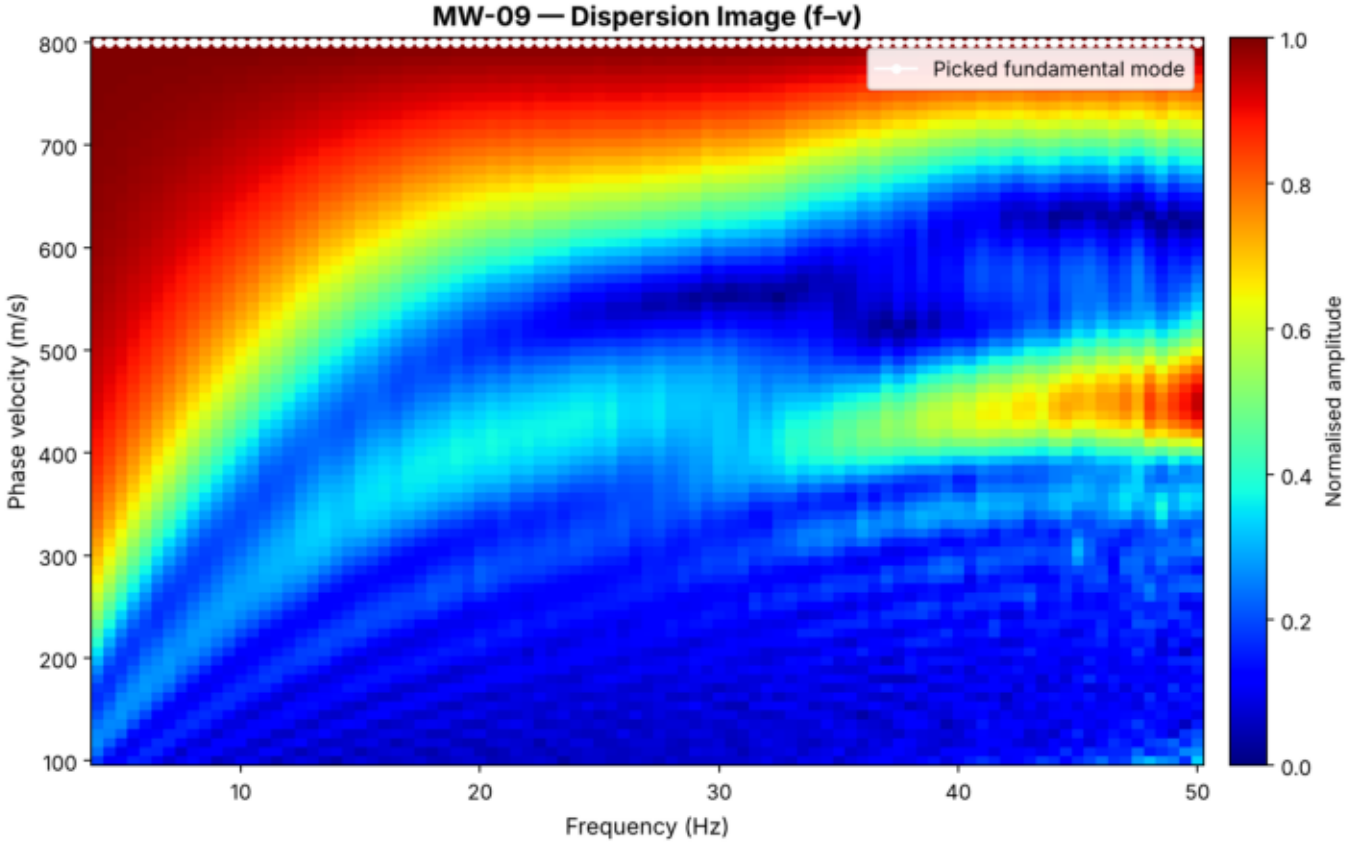
Phase velocity vs frequency, auto-picked fundamental mode.

MW-08 — Vs(z) Profile (BH8, Group G1)



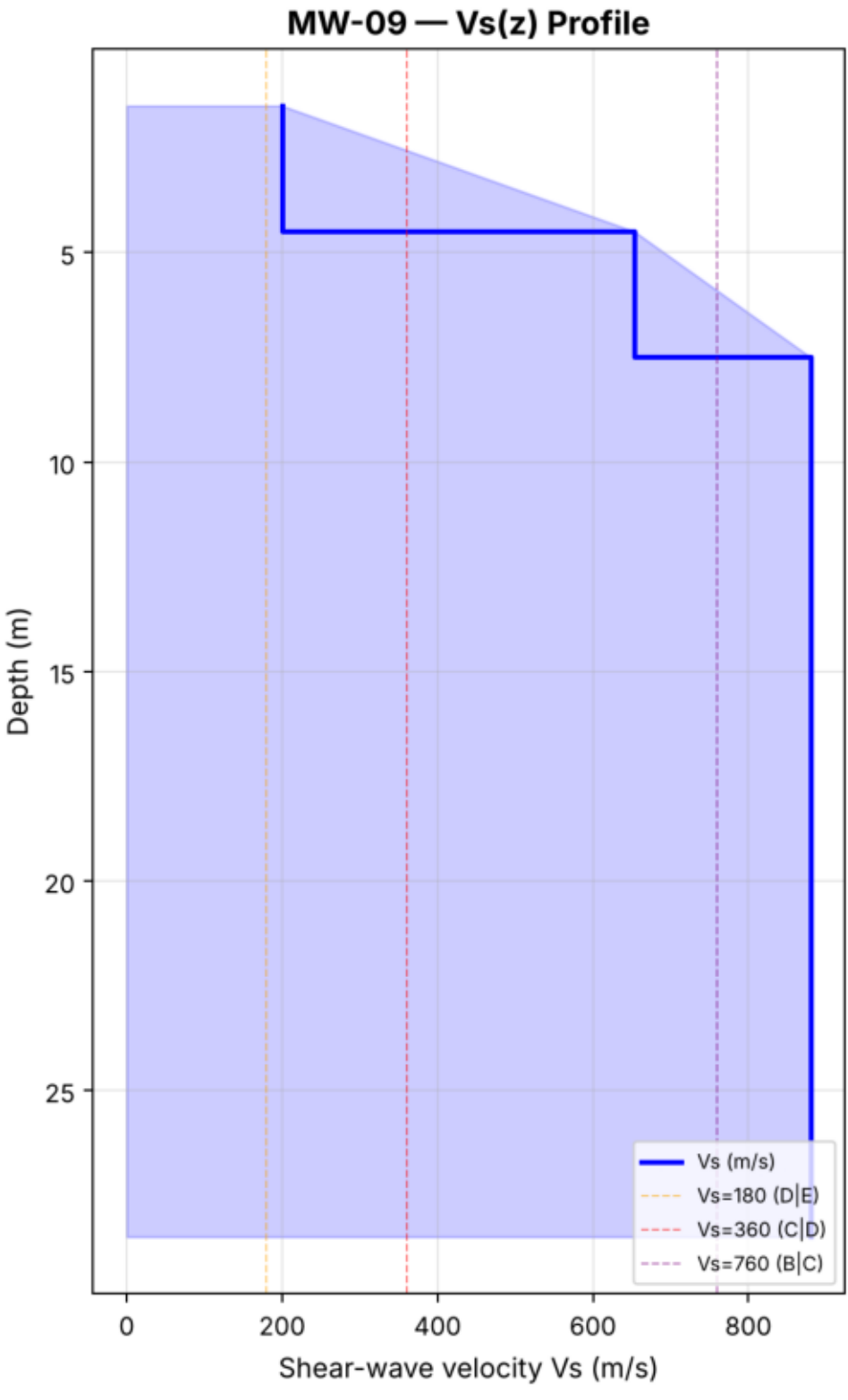
Dispersion inverted to layered Vs profile; a national seismic design code site-class lines overlaid.

MW-09 — Dispersion Image (BH9, Group G2)



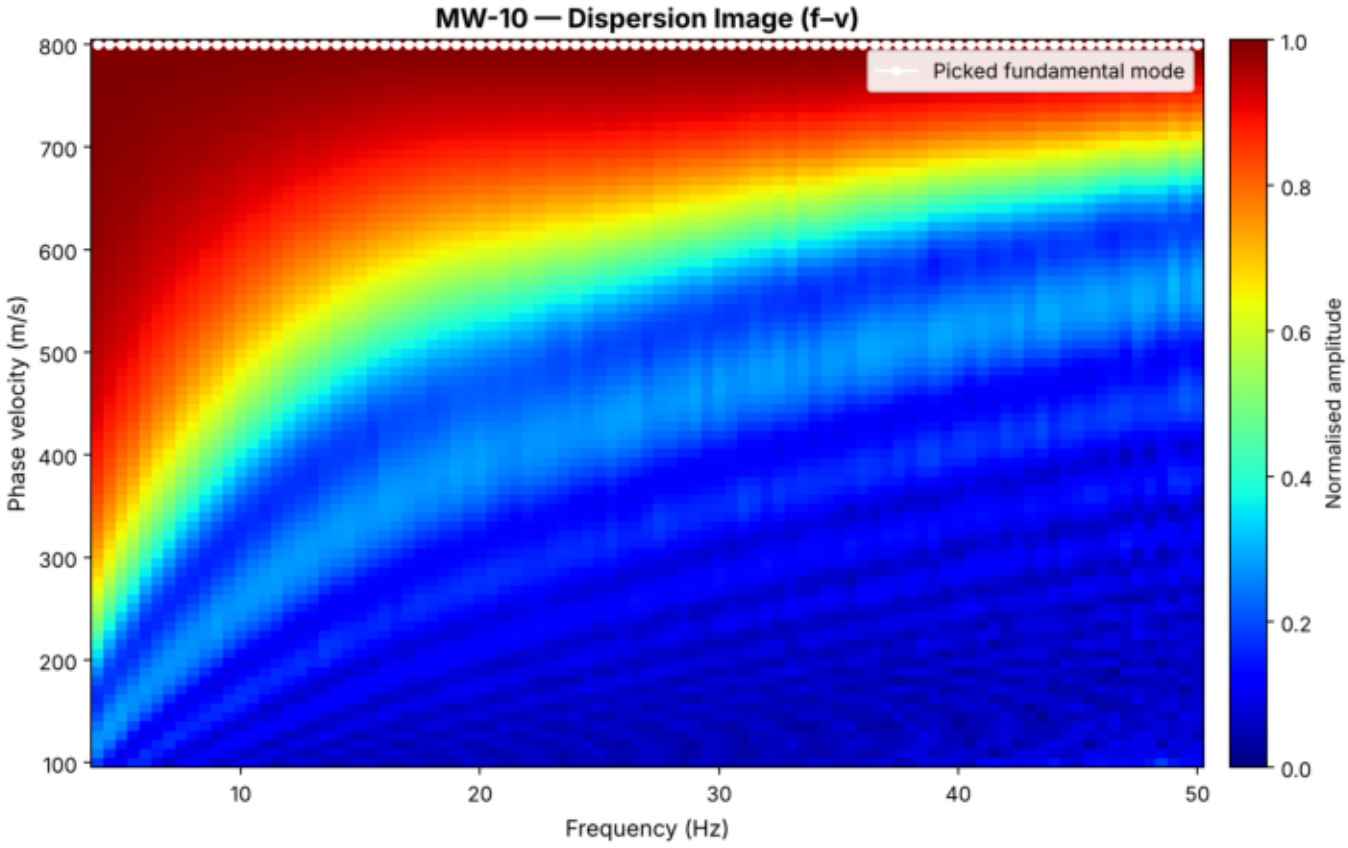
Phase velocity vs frequency, auto-picked fundamental mode.

MW-09 — Vs(z) Profile (BH9, Group G2)



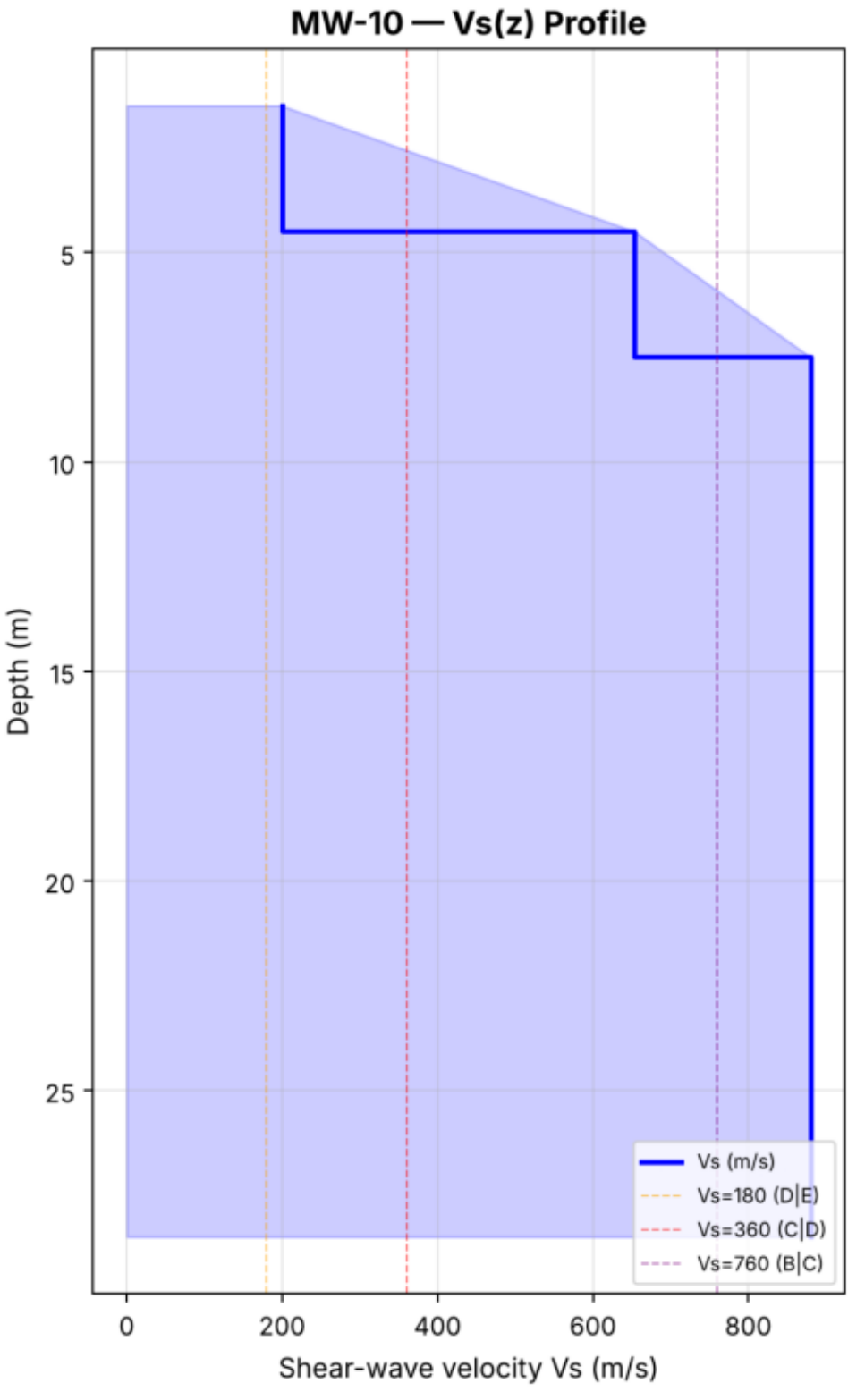
Dispersion inverted to layered Vs profile; a national seismic design code site-class lines overlaid.

MW-10 — Dispersion Image (BH10, Group G2)



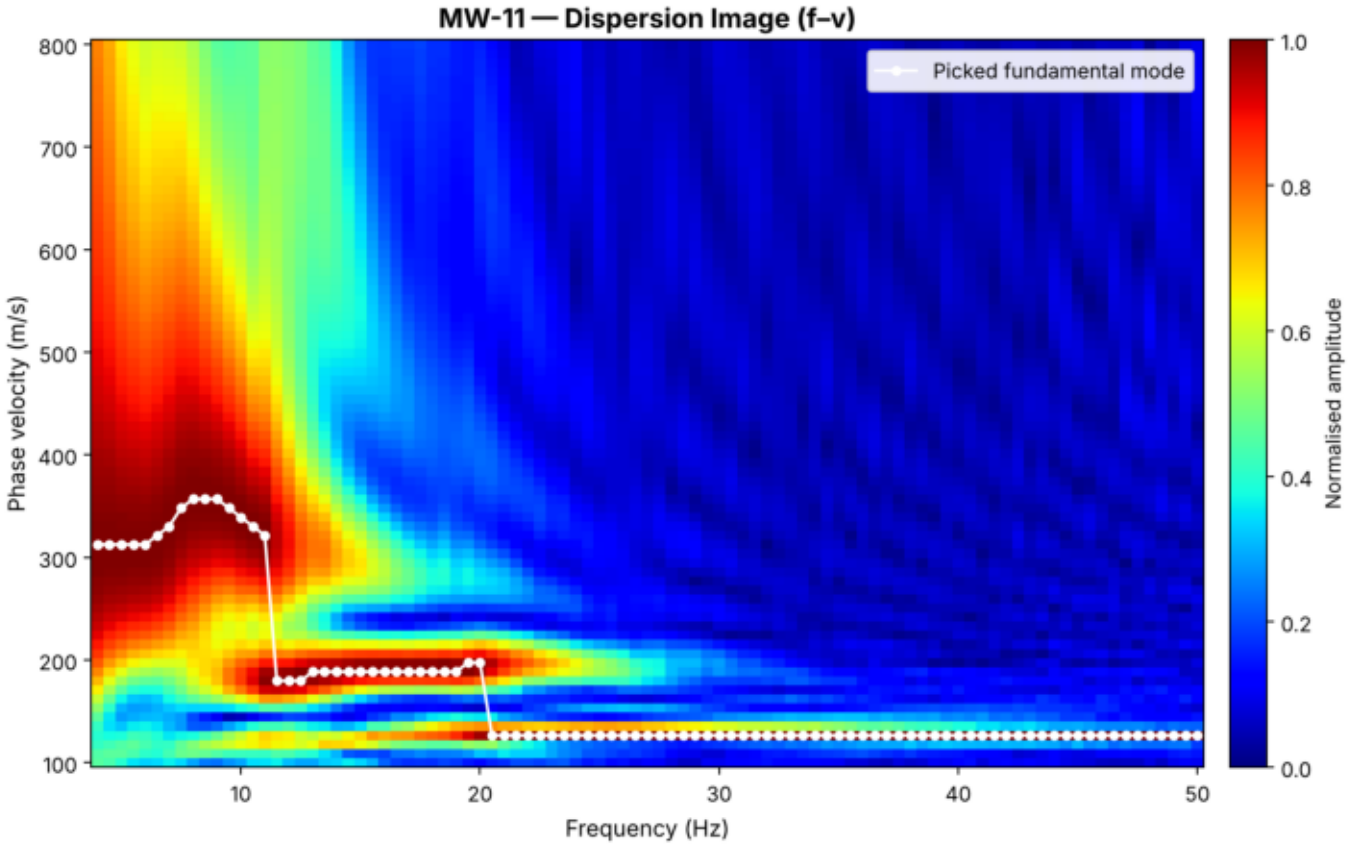
Phase velocity vs frequency, auto-picked fundamental mode.

MW-10 — Vs(z) Profile (BH10, Group G2)



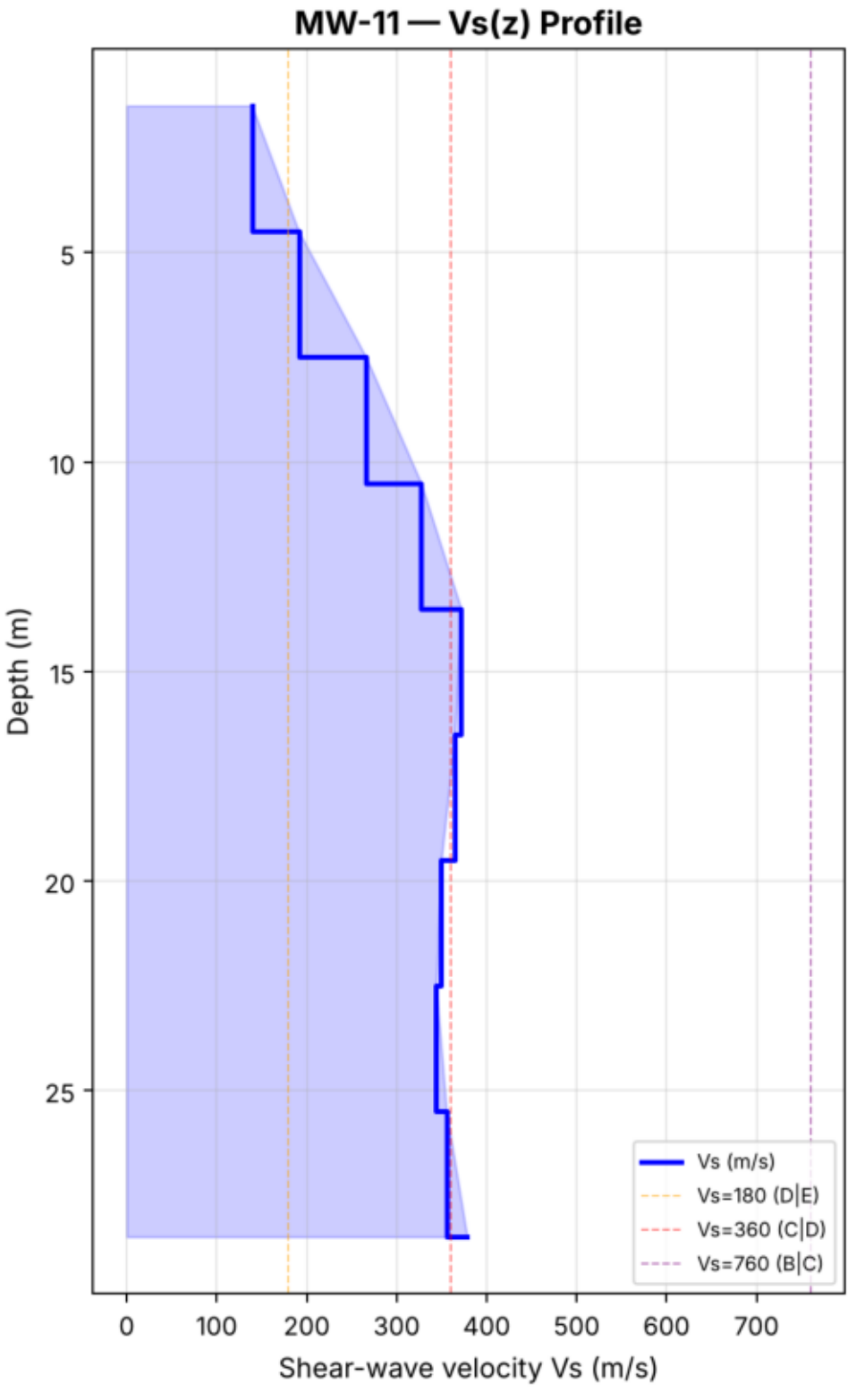
Dispersion inverted to layered Vs profile; a national seismic design code site-class lines overlaid.

MW-11 — Dispersion Image (BH11, Group G5)



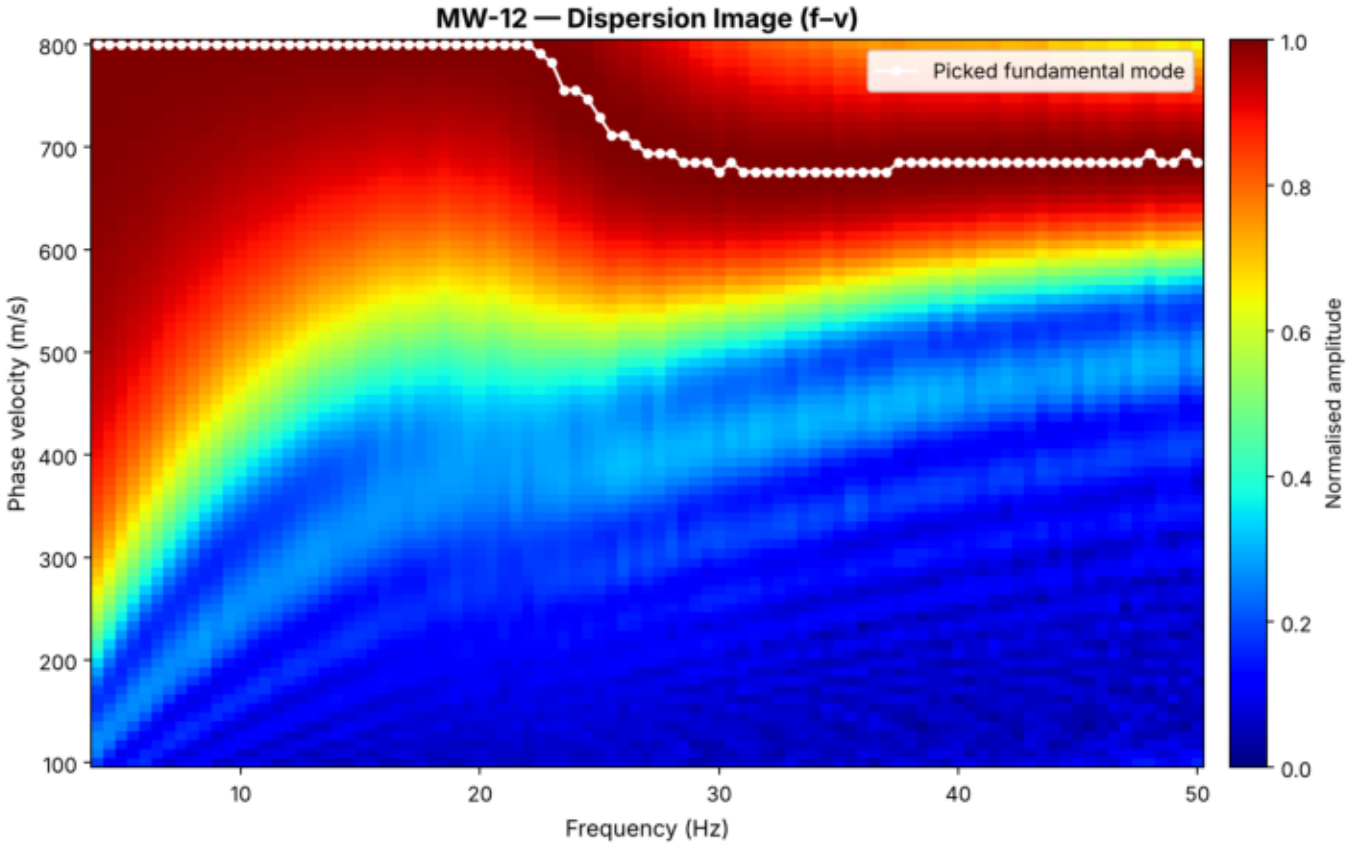
Phase velocity vs frequency, auto-picked fundamental mode.

MW-11 — Vs(z) Profile (BH11, Group G5)



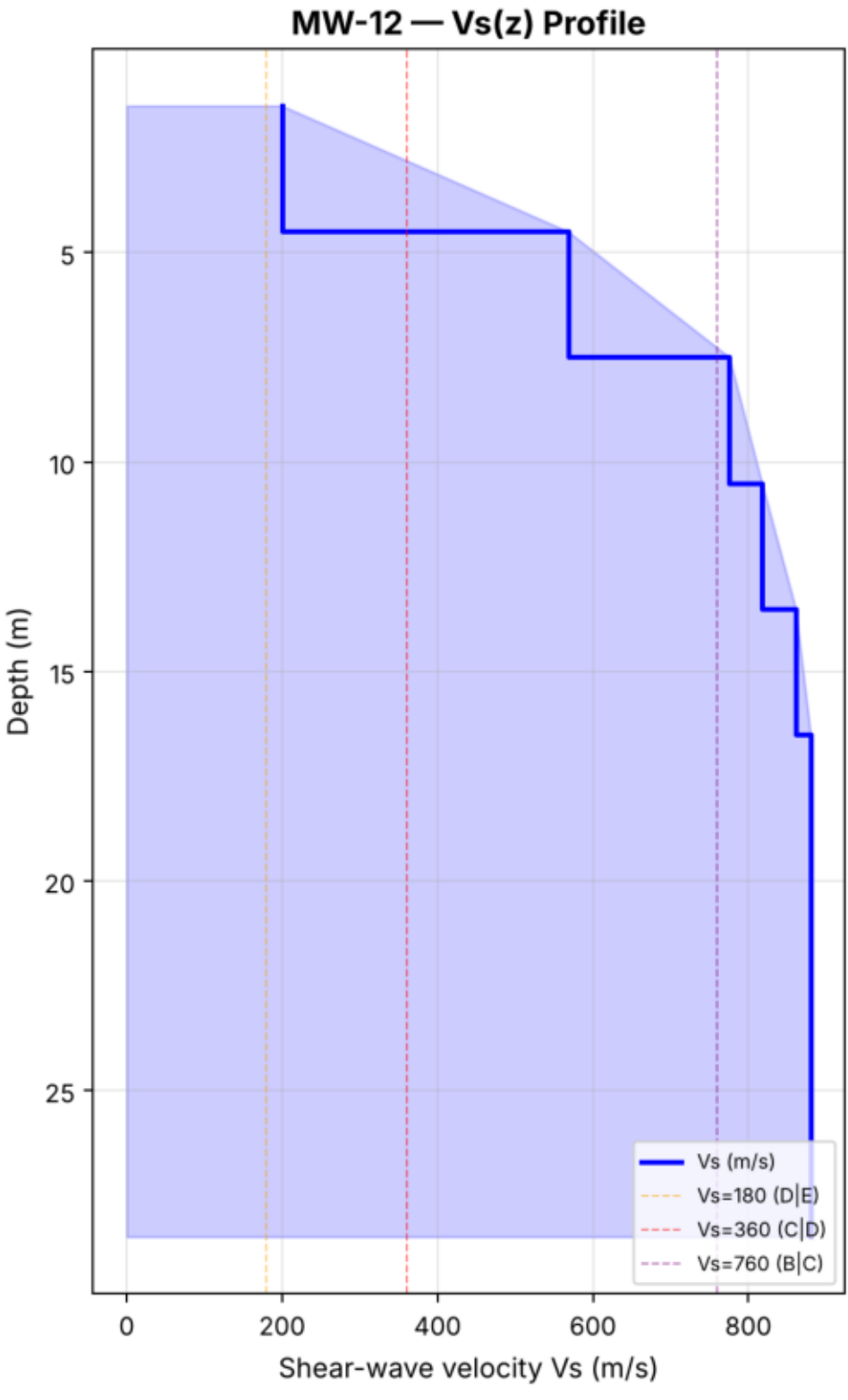
Dispersion inverted to layered Vs profile; a national seismic design code site-class lines overlaid.

MW-12 — Dispersion Image (BH12, Group G1)



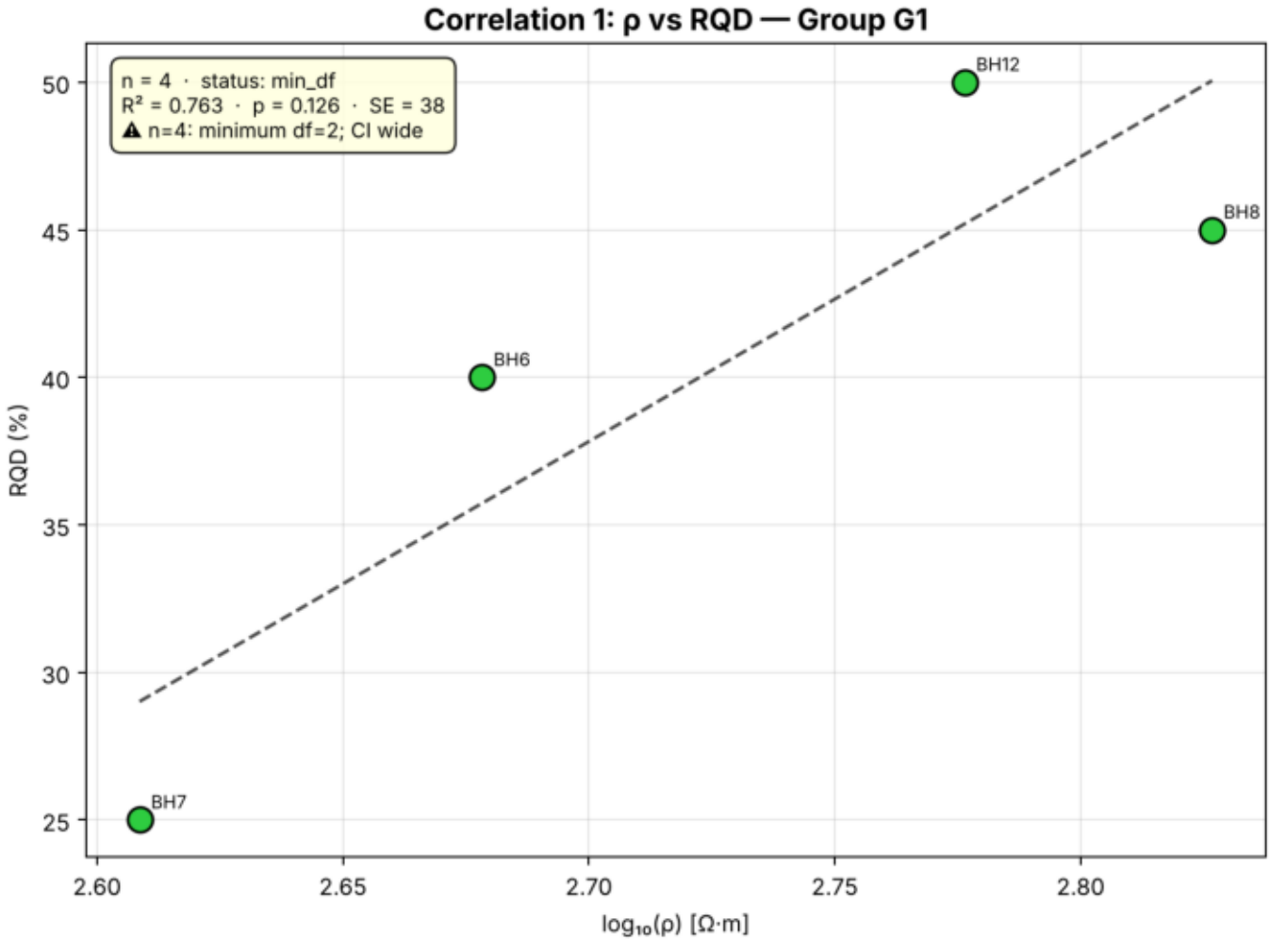
Phase velocity vs frequency, auto-picked fundamental mode.

MW-12 — Vs(z) Profile (BH12, Group G1)

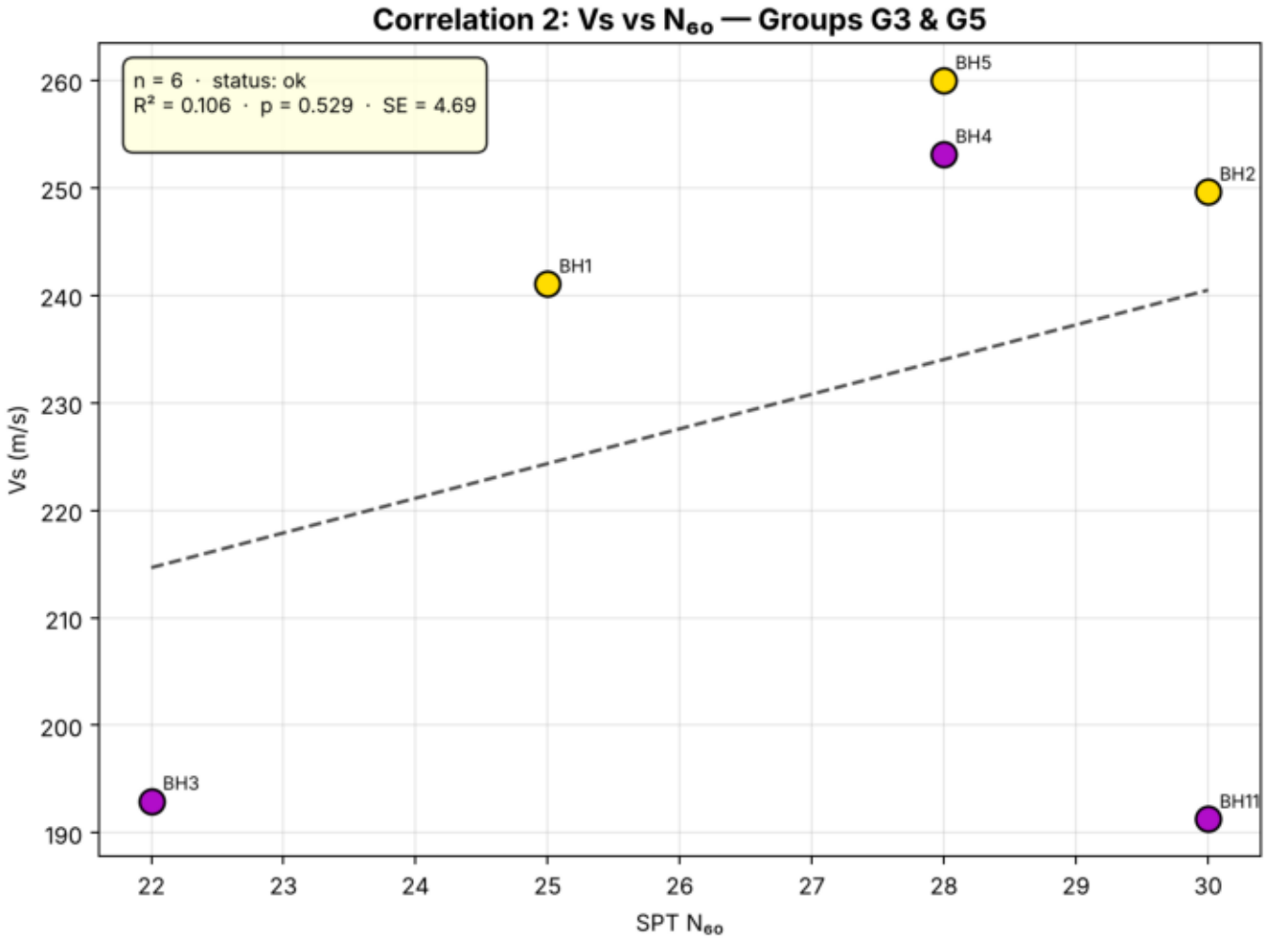


Dispersion inverted to layered Vs profile; a national seismic design code site-class lines overlaid.

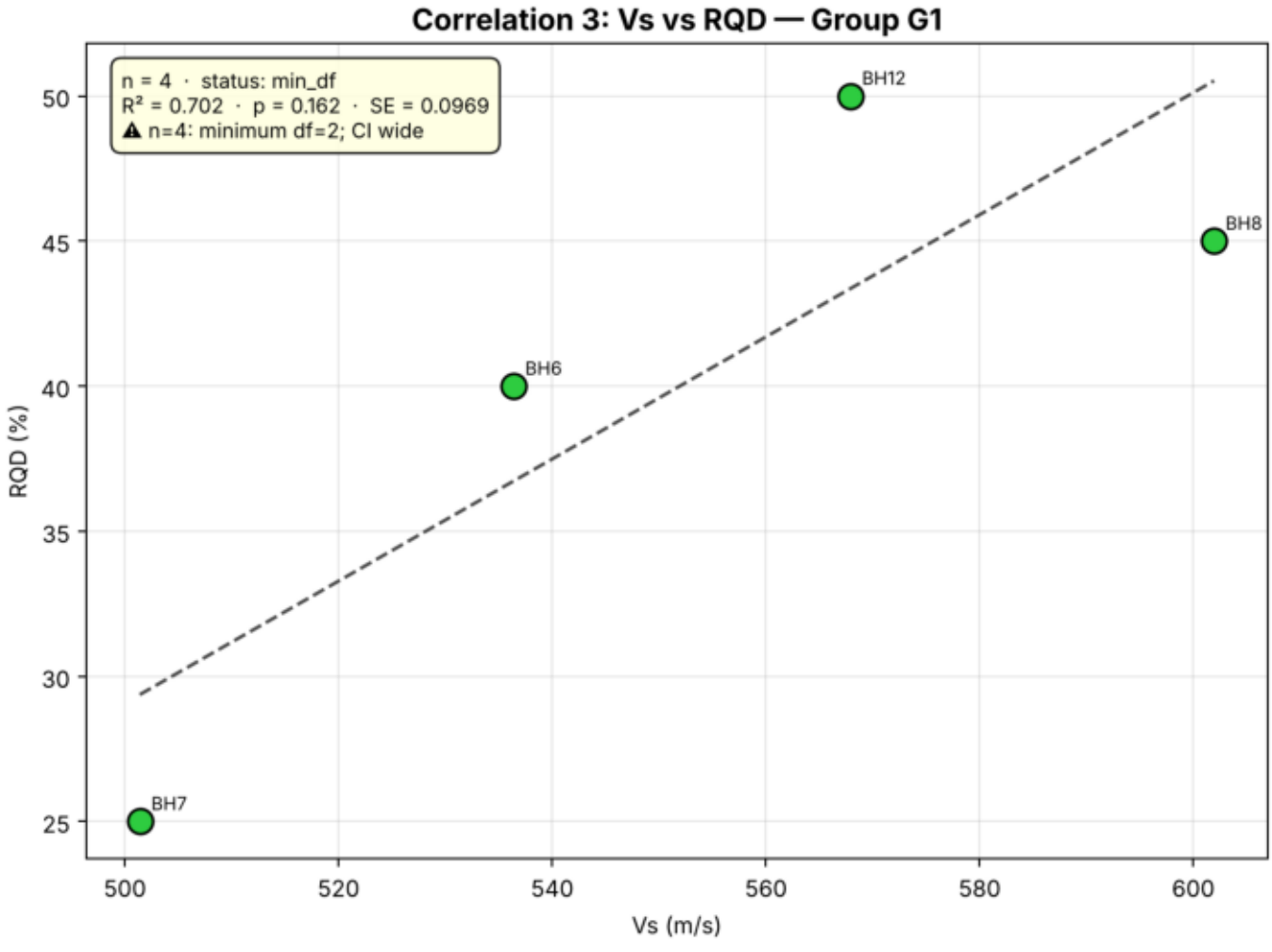
Correlation — corr_1_rho_vs_RQD_G1.png



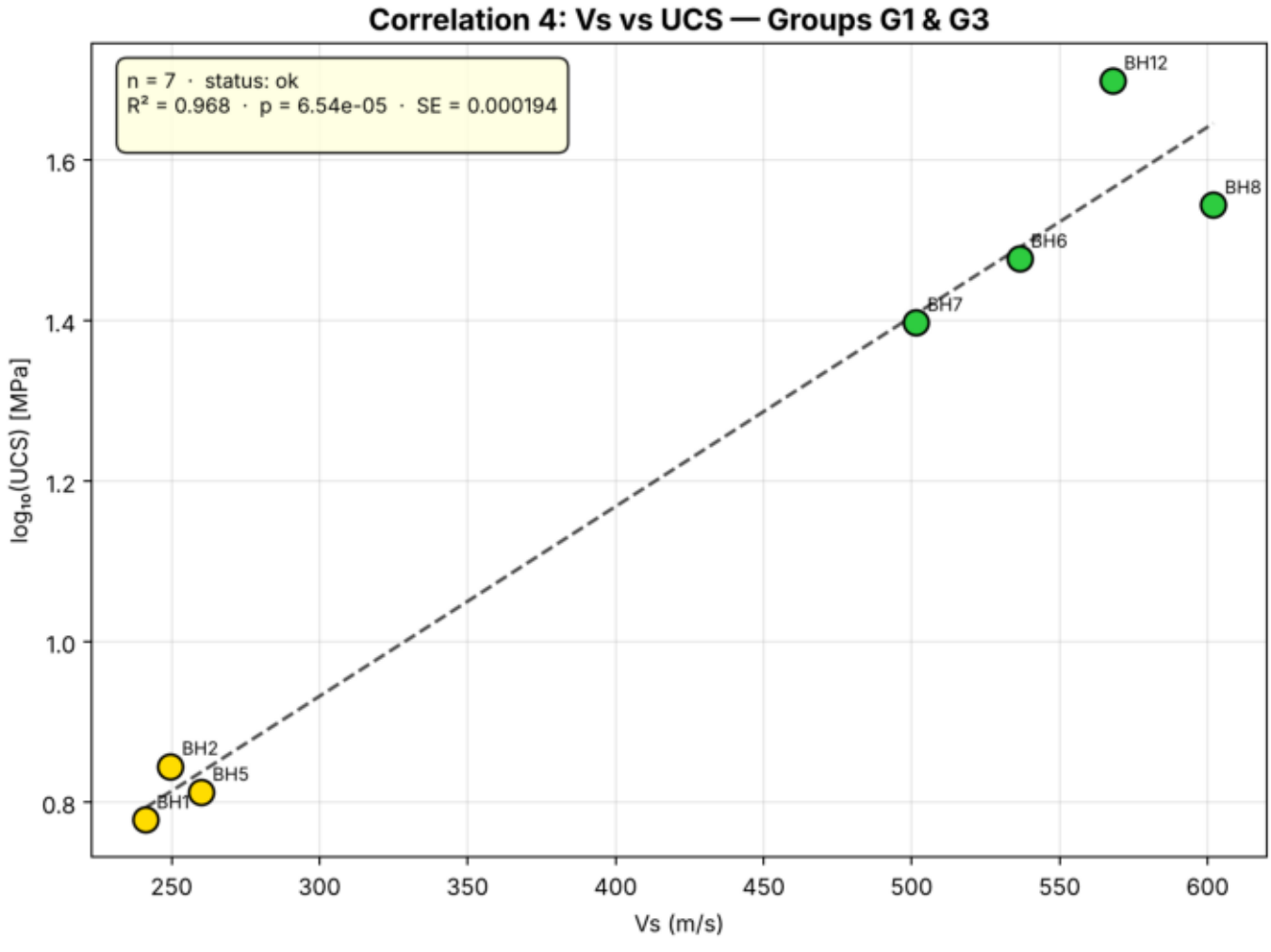
Correlation — corr_2_Vs_vs_N60.png



Correlation — corr_3_Vs_vs_RQD_G1.png



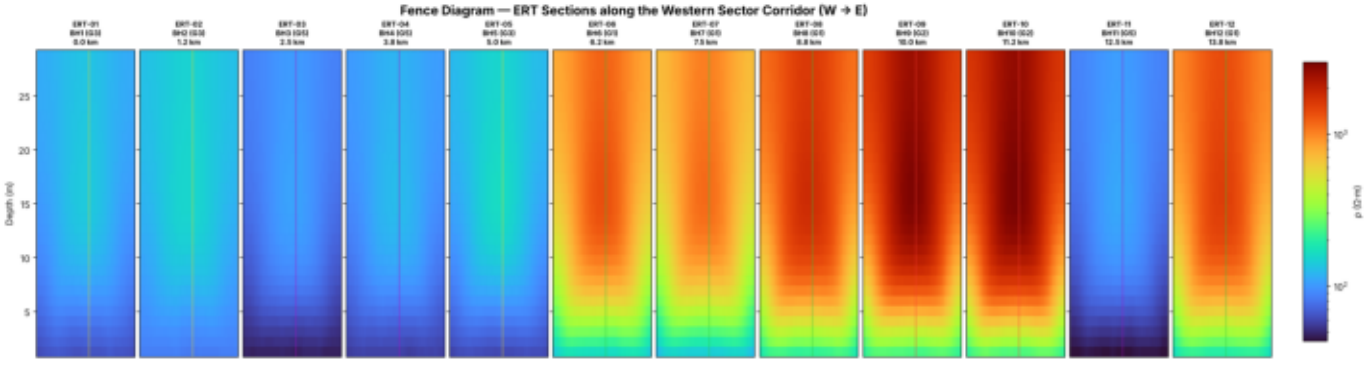
Correlation — corr_4_Vs_vs_UCS.png



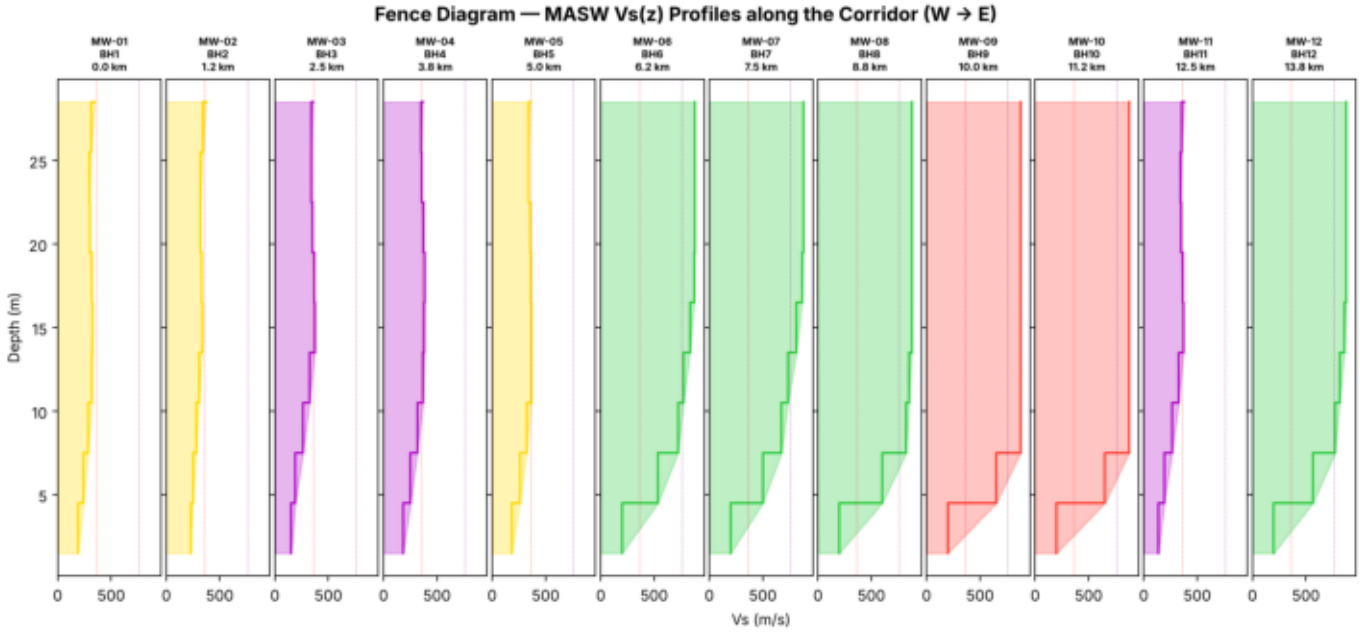
Calibration Data at Target Depth (12 Collocated Sites)

bh	group	rho_at_depth	vs_at_depth	rqd	ucs_mpa	n60	w_grade_num
BH1	G3	88.786	241.071	30.0	6.0	25	5
BH2	G3	104.72	249.639	30.0	7.0	30	4
BH3	G5	67.748	192.951	nan	nan	22	6
BH4	G5	79.175	253.14	nan	nan	28	6
BH5	G3	92.725	259.994	30.0	6.5	28	4
BH6	G1	476.764	536.391	40.0	30.0	60	3
BH7	G1	406.102	501.389	25.0	25.0	60	4
BH8	G1	671.017	601.936	45.0	35.0	60	2
BH9	G2	813.651	653.333	65.0	75.0	60	2
BH10	G2	886.399	653.333	60.0	60.0	60	3
BH11	G5	66.682	191.262	nan	nan	30	6
BH12	G1	597.627	567.913	50.0	50.0	60	2

Fence Diagram — ERT Sections (W → E along corridor)



Fence Diagram — MASW Vs(z) Profiles



Corridor Resistivity Ribbon (interpolated)

